

Mean-Field Peeling

Theory, Algorithm, and Audited Feature-Learning Calculations
Explicit Gaussian Normal Forms for Derivative Observables

Amir Joudaki's mean-field peeling research program

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Source convention. The mathematical content is maintained in two current Markdown sources: the theory and the detailed μP case study. Frozen historical notes are appended for provenance only. This TeX file contains presentation and inclusion logic only, and the current Markdown takes precedence over an older generated PDF. Exact identities, assumption-dependent mean-field results, formal closures, and open theorem targets are distinguished in the source text.

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Part I. Canonical Theory and Theorem Program

Mean-Field Peeling: Theory, Algorithm, and Theorem Program

Status: Canonical maintained theory document

Detailed execution: MUP_TRAINING_CASE_STUDY.md

Exact quadratic specialization: quadratic_compiler/ **Historical source:** archive/ORIGINAL_NOTES_AND_NTK_SAMPLE.md

Last consolidated: 2026-08-13

1. Purpose and status

Mean-field peeling is a layer-by-layer method for reducing a width-normalized scalar contraction of neural-network derivatives to explicit Gaussian expectations. One exposes the highest active layer, conditions on all lower layers, integrates the current Gaussian weights exactly by Wick and Stein identities, passes every lower-layer factor created by that integration to the next layer, and repeats.

This is not an independence argument. A matrix can create both a preactivation and backward or differentiated fields. Their dependence is exactly what generates Stein, or Onsager, response terms. Some response terms vanish after complete width counting; others survive and encode feature learning.

Four claim levels are used throughout.

1. **Exact finite-width identity.** Ordinary differentiation, conditioning, Gaussian integration by parts, or Taylor expansion proves the statement before taking a width limit.
2. **Mean-field result under explicit assumptions.** The displayed limit follows from the peel provided the stated smoothness, moment, leave-finitely-many-out, joint-CLT, concentration, and uniform-integrability assumptions hold. These assumptions are named rather than hidden.
3. **Formal mean-field closure.** The differentiated state and its local Gaussian operations have been specified, but a complete joint convergence theorem for every registered channel has not been supplied.
4. **Open theorem program.** The broad Gaussian-normal-form and depth-linear finite-state claims remain targets, not established theorems.

There is now one important exact specialization. For the one-sample, two-hidden-layer quadratic model, derivative monomials close under a finite decorated-forest grammar. The resulting compiler and its exact order-eleven certificates are described in Section 11. This proves a substantial special-case peeling calculus; it does not upgrade the general theorem target.

For a fixed batch of size B , a primitive preactivation row is a B -dimensional Gaussian vector. Higher-order states may require q jointly generated base vectors and therefore a Gaussian integral of total dimension qB . The number q is finite for each fixed finite construction and is independent of width. A bound on q independent of depth is an additional open finite-state-closure problem.

2. Network, raw parameters, and scaling ledger

Fix a batch

$$\{(x_a, y_a) : a = 1, \dots, B\}.$$

For a fully connected network, write the effective forward weights as

$$A_{ij}^\ell = s_{\ell,n} \theta_{ij}^\ell,$$

where θ^ℓ is the raw parameter optimized during training and $s_{\ell,n}$ is its forward multiplier. The forward pass is

$$z_i^\ell(a) = \sum_j A_{ij}^\ell h_j^{\ell-1}(a) + \sigma_{b,\ell} b_i^\ell, \quad h_i^\ell(a) = \phi_\ell(z_i^\ell(a)),$$

with $h^0(a) = x_a$. Under fan-in Gaussian initialization,

$$A_{ij}^\ell \sim N\left(0, \frac{\sigma_{w,\ell}^2}{n_{\ell-1}}\right).$$

Before any peeling calculation, record:

- every width and limiting width ratio;
- each raw parameter and the corresponding effective forward tensor;
- every weight and bias variance;
- output dimension and readout normalization;
- the layerwise optimizer or μP metric factor;
- the observable normalization and desired asymptotic width order;
- whether the target is an expectation, a deterministic probability limit, or a fluctuation law.

No width power counting is meaningful until this ledger is fixed.

3. Exact layerwise conditional Gaussian law

Let

$$\mathcal{F}_{\ell-1} = \sigma(A^1, b^1, \dots, A^{\ell-1}, b^{\ell-1}).$$

Conditional on $\mathcal{F}_{\ell-1}$, the row blocks

$$(A_{i,\cdot}^\ell, b_i^\ell, Z_i^\ell), \quad Z_i^\ell = (z_i^\ell(1), \dots, z_i^\ell(B)),$$

are independent centered jointly Gaussian vectors across i . Exactly at finite width,

$$Q_{n,ab}^\ell := \text{Cov}(z_i^\ell(a), z_i^\ell(b) \mid \mathcal{F}_{\ell-1}) = \frac{\sigma_{w,\ell}^2}{n_{\ell-1}} \sum_j h_j^{\ell-1}(a) h_j^{\ell-1}(b) + \sigma_{b,\ell}^2, \quad (3.1)$$

$$\text{Cov}(A_{ij}^\ell, A_{i'j'}^\ell) = \delta_{ii'} \delta_{jj'} \frac{\sigma_{w,\ell}^2}{n_{\ell-1}}, \quad (3.2)$$

and

$$\text{Cov}(A_{ij}^\ell, z_{i'}^\ell(a) \mid \mathcal{F}_{\ell-1}) = \delta_{ii'} \frac{\sigma_{w,\ell}^2}{n_{\ell-1}} h_j^{\ell-1}(a). \quad (3.3)$$

Equation (3.3) is the source of the Stein response. It contains a lower-layer activation, so eliminating group ℓ creates boundary data for group $\ell - 1$.

Unconditionally, the covariance in (3.1) is random. Deep variables across several layers therefore form a Gaussian mixture in general, not one global joint Gaussian vector. Peeling relies on exact conditional Gaussianity one layer at a time.

4. Deterministic forward state

Under the required law-of-large-numbers and moment assumptions,

$$Q_n^\ell \longrightarrow Q^\ell.$$

The deterministic recursion is

$$Q_{ab}^1 = \sigma_{w,1}^2 \frac{x_a^\top x_b}{n_0} + \sigma_{b,1}^2,$$

and

$$Q_{ab}^{\ell+1} = \sigma_{w,\ell+1}^2 \mathbb{E}_{Z \sim N(0, Q^\ell)} [\phi_\ell(Z_a) \phi_\ell(Z_b)] + \sigma_{b,\ell+1}^2. \quad (4.1)$$

Labels do not enter these initialization covariances. They enter through loss derivatives and training-time vector fields.

5. Groups, explicit atoms, and admissible observables

Group ℓ contains:

- raw or effective layer- ℓ parameter coordinates and biases;
- preactivation coordinates $z_i^\ell(a)$;
- scalar functions of those coordinates, including $\phi_\ell, \phi'_\ell, \phi''_\ell, \dots$.

A backward error, Hessian-vector product, or training-time derivative is not initially explicit. Chain rule, product rule, and Faà di Bruno expansion turn each fixed-order derivative into a finite sum of products of explicit atoms.

The proposed admissible input language consists of scalar expressions generated from:

- outputs and losses on the fixed batch;
- preactivations, activations, and hidden-Gram entries;
- parameter, input, or training-time derivatives of total order at most a fixed R ;
- finite sums, products, and tensor contractions;

subject to the following structural conditions:

1. all width-growing neuron and parameter indices are eventually contracted;
2. every width normalization is declared;
3. the local expression schema is independent of width;
4. depth dependence is generated by repeated layer-local templates;
5. the expression is width-balanced under the declared scaling.

This definition is intentionally syntactic. “Any observable having a mean-field limit” would be circular and too broad. A width-growing uncontracted Hessian is not one scalar peeling state. Ratios, inverses, maxima, spectral edges, and similar global nonlinear functions require additional nondegeneracy or regularity arguments.

6. Scalar contraction normal form

After derivative expansion and scalarization, one term has the form

$$T_n = c_n \sum_i \mathbf{1}\{\mathcal{C}(i)\} \prod_{r=1}^{R_W} A_{i_r j_r}^{\ell_r} \prod_{s=1}^{R_Z} \psi_s(z_{u_s}^{k_s}(a_s)). \quad (6.1)$$

The contraction state records:

- every neuron index and its layer;
- all fixed batch labels;
- exact equality and inequality constraints;
- the derivative order of every scalar function;
- all explicit width powers;
- every open lower-layer index passed to the next peel.

Once indices are exposed, every factor is scalar and may be reordered by group. Recursive scalarization exposes only the current highest group rather than expanding the whole depth at once.

The desired final output is a finite Gaussian-normal-form directed acyclic graph. A Gaussian atom has the form

$$\mathcal{J}_{\ell,F} = \int_{\mathbb{R}^d} F(g) N(0, K^\ell)(dg), \quad (6.2)$$

with the layer ℓ , covariance K^ℓ , dimension d , integrand F , all activation derivatives, deterministic coefficients, and surviving batch-index/equality contractions explicitly specified. The final output contains no random weights, neuron-index sums, implicit backward vectors, recursively named

limit variables, or unevaluated Onsager terms.

7. Exact Wick–Stein elimination

7.1 One weight

If (X, Z) is centered jointly Gaussian and F is differentiable with integrable derivative, then

$$\mathbb{E}[XF(Z)] = \sum_{\alpha} \text{Cov}(X, Z_{\alpha}) \mathbb{E}[\partial_{\alpha} F(Z)]. \quad (7.1)$$

For a current-layer weight, conditional on the lower filtration,

$$\mathbb{E}[A_{ij}^{\ell} F(Z_i^{\ell}) \mid \mathcal{F}_{\ell-1}] = \frac{\sigma_{w,\ell}^2}{n_{\ell-1}} \sum_{a=1}^B h_j^{\ell-1}(a) \mathbb{E}[\partial_{z_i^{\ell}(a)} F(Z_i^{\ell}) \mid \mathcal{F}_{\ell-1}]. \quad (7.2)$$

The explicit weight disappears, a derivative is added to the nonlinear factor, and the covariance inserts a lower-layer activation. The latter must remain explicit until its own group is peeled.

7.2 Two weights

For centered jointly Gaussian X, Y, Z ,

$$\begin{aligned} \mathbb{E}[XYF(Z)] &= \text{Cov}(X, Y) \mathbb{E}[F(Z)] \\ &+ \sum_{\alpha, \beta} \text{Cov}(X, Z_{\alpha}) \text{Cov}(Y, Z_{\beta}) \mathbb{E}[\partial_{\alpha\beta} F(Z)]. \end{aligned} \quad (7.3)$$

The first line is the weight–weight Wick branch. The second is the double Stein, or Onsager, branch. Neither may be discarded before global counting.

7.3 General partial matchings

Let $X_1, \dots, X_r$ be explicit Gaussian parameter coordinates in the current group and let $F(Z)$ contain all current preactivation functions. Conditional on lower groups,

$$\begin{aligned} \mathbb{E} \left[\prod_{q=1}^r X_q F(Z) \right] &= \sum_{\pi} \left(\prod_{\{q,t\} \in \pi} \text{Cov}(X_q, X_t) \right) \\ &\times \sum_{\alpha: U(\pi) \rightarrow \mathcal{J}_Z} \left(\prod_{q \in U(\pi)} \text{Cov}(X_q, Z_{\alpha(q)}) \right) \mathbb{E}[\partial_{\alpha(U(\pi))} F(Z)]. \end{aligned} \quad (7.4)$$

Here π ranges over partial matchings of the weight occurrences and $U(\pi)$ is the unmatched set. A matched pair is a weight–weight Wick contraction. An unmatched occurrence attaches to a preactivation through Stein’s identity. Every attachment differentiates the top nonlinear factor and inserts the corresponding lower-layer activation.

Complete matchings are sometimes called pseudo-independent branches; partial matchings are Onsager branches. The phrase “independent copy plus correction” is only a mnemonic. Equation (7.4) is the actual identity, and branch names do not determine asymptotic size.

8. Equality partitions and global width degree

For top-row indices $i_1, \dots, i_q$, split the sum by exact equality partition:

$$\sum_{i_1, \dots, i_q} = \sum_{\pi \in \mathcal{P}_q} \sum_{\ker(i) = \pi}. \quad (8.1)$$

A partition with $|\pi|$ distinct blocks has

$$n_{\ell}^{|\pi|} (1 + O(n_{\ell}^{-1}))$$

assignments before other constraints are imposed. For one row block, define

$$\Gamma_{\ell}[(a_1, \psi_1), \dots, (a_s, \psi_s); Q] := \mathbb{E}_{Z \sim N(0, Q)} \left[\prod_{r=1}^s \psi_r(Z_{a_r}) \right]. \quad (8.2)$$

Different row blocks yield products of such moments, subject to any joint base-channel structure retained by the state.

For iid rows Z_i , the elementary two-index example is

$$\frac{1}{n^2} \sum_{i,j} F(Z_i) G(Z_j) = \frac{1}{n^2} \sum_{i \neq j} F(Z_i) G(Z_j) + \frac{1}{n^2} \sum_i F(Z_i) G(Z_i). \quad (8.3)$$

The off-diagonal partition has two free row blocks and converges to $\mathbb{E}[F(Z)] \mathbb{E}[G(Z)]$. The diagonal partition has one free row block and is lower by n^{-1} under this normalization. If the observable instead contains $n^{-1} \sum_i F(Z_i) G(Z_i)$, that diagonal joint moment is leading. This is the smallest example of why equality patterns and normalization must be considered together.

The global width degree includes:

- one positive width power for each free neuron-index block;
- negative powers from weight covariances and weight–preactivation attachments;
- the observable normalization;

- raw-to-effective parameter multipliers;
- optimizer or μP metric factors;
- every lower-layer sum created by a Stein attachment.

A branch may be dropped only after its full lower-layer boundary signature is known. A locally subleading branch can create another lower-layer sum and recover its order. A locally comparable branch can later be forced onto a diagonal and lose an order. Likewise, an off-diagonal family with zero one-copy mean can survive in a two-copy calculation as a fresh Gaussian fluctuation field.

9. Deterministic covariance replacement

Exact conditional integration initially produces Gaussian moments at the lower-random empirical covariance Q_n^ℓ . Only after distinguished rows and equality patterns have been exposed may one use

$$Q_n^\ell \longrightarrow Q^\ell$$

and conclude, under continuity and uniform integrability,

$$\Gamma_\ell[\dots; Q_n^\ell] \longrightarrow \Gamma_\ell[\dots; Q^\ell]. \quad (9.1)$$

If a distinguished row also enters Q_n^ℓ , a quantitative proof first uses a leave-one-out or leave-finitely-many-out covariance. Removing finitely many rows changes the covariance by the relevant small width order and makes the remaining covariance independent of those distinguished rows.

For a centered or fluctuation-amplified target, the zeroth-order replacement can erase the leading term. One must then expand in $Q_n^\ell - Q^\ell$. Price's identity keeps this expansion inside the same Gaussian derivative algebra:

$$D \mathbb{E}_{Z \sim N(0, Q)}[F(Z)][H] = \frac{1}{2} \sum_{a,b} H_{ab} \mathbb{E}[\partial_{ab} F(Z)], \quad (9.2)$$

with the usual symmetric-covariance convention.

10. The peeling algorithm

10.1 Inputs

Specify the architecture, widths, activations, raw/effective parameter scalings, output and bias normalization, optimizer metric, fixed batch, scalar observable, maximum derivative order, requested width order, and convergence mode.

10.2 One maximal-layer elimination

1. Expand derivative rules only far enough to expose the highest active group.
2. Scalarize all contractions involving that group.
3. Record exact width powers, equality constraints, and open lower-layer indices.
4. Reorder scalar atoms by group and select the highest active group p .
5. Condition on $\mathcal{F}_{p-1}$.
6. Split top-row indices into compatible exact equality partitions.
7. Apply (7.4) until no explicit group- p Gaussian parameter remains.
8. Expand every derivative created on a nonlinear factor.
9. Substitute the exact conditional covariances.
10. Resolve Kronecker deltas and update the lower-layer boundary state.
11. Count widths globally, including every downstream sum.
12. Convert surviving top-row functions to Gaussian row moments at the empirical covariance Q_n^p .
13. Replace Q_n^p by Q^p , or retain the covariance fluctuation required by the target.
14. Canonicalize and merge algebraically identical lower states.
15. Repeat at group $p - 1$.
16. At layer zero, evaluate deterministic input and label contractions.

Exact conditional elimination and deterministic mean-field replacement are separate steps. An index may be summed during group p only when none of its remaining occurrences belongs to a lower group.

10.3 Convergence pass

If a deterministic limit for a typical initialization is claimed, repeat the calculus on the square, covariance, or relevant cumulants. A one-copy expectation calculation does not prove concentration.

10.4 Recursive-state form

If the boundary signatures close on a finite state family $\mathcal{S}$, one peel has the form

$$V_p(s) = \sum_{s' \in \mathcal{S}} T_p(s, s'; Q^p) V_{p-1}(s'), \quad (10.1)$$

and hence

$$V_L = T_L(Q^L) \dots T_1(Q^1) V_0. \quad (10.2)$$

The conservative statement is that maximal-layer elimination terminates after at most L layer stages. An $O(L)$ -size shared DAG additionally requires a state-space bound independent of depth at fixed batch size, derivative order, step count, and observable schema. That finite-state strengthening remains open in general.

11. Exact quadratic decorated-forest specialization

11.1 Model and scope

At activation scale $\gamma = 1$, the one-sample, two-hidden-layer quadratic model collapses to

$$z_i = \frac{1}{\sqrt{n}} \sum_{j=1}^n W_{ij} u_j^2, \quad f_n = \frac{1}{n} \sum_{i=1}^n a_i z_i^2 = \frac{1}{n^2} \sum_{i,j,k} a_i W_{ij} W_{ik} u_j^2 u_k^2,$$

with all a_i, W_{ij}, u_j independent standard Gaussians. Feature-ascent differentiation of an observable A is

$$D_n A = n \langle \nabla f_n, \nabla A \rangle,$$

where the gradient includes every a, W, u coordinate. For fixed k ,

$$J_k = \lim_{n \rightarrow \infty} \mathbb{E}[D_n^k f_n].$$

If F denotes the formal feature jet, then $F^{(k)}(0) = J_k$. The derivative order is fixed before $n \rightarrow \infty$; no uniform-in- k bound, positive-time trajectory, or convergence of the formal series follows. The readout flip and time reversal give $J_{2r} = 0$.

11.2 Exact scalar graph grammar

Every scalar monomial generated by repeated differentiation is a decorated bipartite forest:

- a row vertex stores a row index and the exponent of a_i ;
- a column vertex stores a column index and half the exponent of u_j ;
- an edge is one occurrence of W_{ij} .

The derivation has exactly three local rewrites.

1. An a -hit adds the two weight edges contributed by z_i^2 , with the current a -multiplicity.
 2. A hit on u_j^{2p} adds the new row/column gadget with coefficient $8p$.
 3. A W_{ij} -hit has coefficient 2, deletes that bridge, changes its endpoint decorations, and adds the fresh edge prescribed by the vector field.
- The first two preserve connectedness. A weight hit splits a tree into two trees. If $k-1$ derivatives remain, the exact Leibniz contribution is

$$2 \sum_{q=0}^{k-1} \binom{k-1}{q} A_q(T_1) A_{k-1-q}(T_2).$$

The sector-graded recurrence also records the number w of weight hits. At total derivative order k , the number of Wick covariance pairs is

$$P = k + 1 - w.$$

11.3 Leading-width factorization theorem

Take a generated normalized forest with r original components and $2P$ weight edges. A Wick pairing gives a quotient covariance graph with P edges. If it has V vertices, c components, and cycle rank β , then

$$V = P + c - \beta.$$

The state normalization makes the contraction leading exactly when $V = P + r$. Pairing can only merge original components, so $c \leq r$, and $\beta \geq 0$. Therefore survival requires

$$c = r, \quad \beta = 0.$$

No leading covariance pair joins two original components, every quotient component is a tree, and the leading expectation factors across the original components. This exact result is scoped to the generated quadratic forest grammar and its normalization; it is not a general independence principle.

11.4 Accepted exact coefficients

The accepted computational certificates are

$$\begin{aligned} F'(0) &= 111, \\ F^{(3)}(0) &= 1\,685\,184, \\ F^{(5)}(0) &= 77\,400\,633\,120, \\ F^{(7)}(0) &= 7\,315\,868\,433\,079\,296, \\ F^{(9)}(0) &= 1\,181\,161\,141\,825\,400\,561\,664, \\ F^{(11)}(0) &= 291\,982\,832\,387\,585\,872\,335\,470\,592. \end{aligned}$$

The order-eleven number is the exact sum of all twelve Wick-pair sectors. The audit combines a transparent Python prototype, direct pairings at low order, exhaustive differentiated-forest expansion, compressed connected recursion, and a second equality-partition evaluator. Every accepted higher-order route first reproduced all ten order-nine sectors exactly. Order-eleven sectors $P = 1, \dots, 9$ were evaluated from exhaustive exports with checked 512-bit arithmetic; $P = 10, 11, 12$ were separately certified by the high-sector machinery. A conservative bound needs only 275 bits through order thirteen, so overflow would throw rather than wrap.

These are exact integer outputs of audited computer algebra under the proved special-case reduction, not hand enumerations or formally verified software. In particular, the order-eleven $P = 9$ sector has one complete checked route but has not been redundantly recomputed by the newer connected-sector engine. Full provenance is in the D11 audit and sector-engine audit.

11.5 Order-thirteen and acceleration audits

The complete $F^{(13)}(0)$ is unknown. Exact positive-subsum calculations give monotone lower bounds, culminating at

$$50\,393\,647\,763\,255\,899\,049\,472\,742\,772\,736$$

for component-edge cap fourteen. This is only a lower bound. In the maximal $P = 14$ sector, discovery found 465,075 recurrence states and 325,190 base trees; only the first 704 base contractions completed, so no sector subtotal was certified. See the D13 audit.

Several accelerations were audited:

- approximate first-pair and adjacent-pair modes retain positive sub-sums but are not exact checkpoints;
- the GF(2) matroid shortcut is valid only in the no-weight-hit, even-row, tight-nullity case; an explicit decorated-tree counterexample invalidates its unrestricted rank gate;
- a singleton-row shortcut omitted even- a parity, so its reported subtotal and timing are retracted;
- prefixwise factor-nine two-hit bounds are exactly false, and an all-order version would contradict the factorial lower bound; no special order-eleven upper bound was proved;
- finite-width formal-jet simulation is a numerical pilot, not an exact mean-field certificate.

The restricted proofs, counterexamples, and retractions live next to the compiler. They illustrate a central MFP rule: a shortcut earns only the scope of its proof, even after passing extensive low-order regression gates.

11.6 Model-specific parameter extensions

Six bounded compiler campaigns tested how much of the same exact grammar survives beyond the canonical point.

1. A relative block-metric ray jointly computed the output through order nine and the second-hidden squared-RMS response through order eight.
2. Two symmetry-reduced inputs with equal and opposite labels were computed through order seven while retaining the input Gram matrix in both the initialization law and the first-layer gradient metric.
3. A centered-to-overcentered first-hidden quadratic activation was computed through order seven using an exact centered-Gaussian moment grammar.
4. Independent first-hidden and middle-weight metric parameters were graded into 125 atomic sectors through order nine on the full nonnegative quadrant.
5. Three equicorrelated equal-label inputs were compiled through order five, including a genuine signed three-color cycle invariant; the order-seven resource gate failed closed.
6. A bounded canonical order-thirteen threshold probe tested coarse and hybrid envelopes but failed its fresh-regression/provenance gate and produced no accepted new bound.

Campaigns 1–4 reached exact finite Hankel endpoints in the downstream Stieltjes study. Campaign 5 proves only exact lower jets and two necessary moment signs, not a Hankel determinant. Campaign 6 is a stopped diagnostic, not a coefficient or bound. These extensions enlarge the audited special-case compiler but do not establish the general MFP theorem. Their consolidated claim levels are in the downstream Stieltjes master.

11.7 What this does and does not establish

The compiler establishes full scalarization, exact derivative rewrites, leading-width Wick selection, equality-identification bookkeeping, component factorization, canonical-state memoization, and exact integer evaluation for this special model.

It does not cover arbitrary depth or batch size, general smooth activations, conditional weight–activation Stein attachments, random empirical-covariance replacement, arbitrary biases or readouts, general muP optimizer metrics, concentration, fluctuation laws, or depth-independent state closure. It is an exact special-case realization and regression laboratory, not a proof of the full Mean-Field Derivative Peeling Theorem.

12. Backward-kernel laboratory: what the first execution establishes

The complete fixed-network calculation is in the case study. Its role in the theory is to expose the simplest nontrivial state transition.

For a centered $1/n$ readout and three hidden layers, define normalized backward variables $\delta_i^\ell(a)$ and empirical kernels

$$\Pi_{n,ab}^\ell = \frac{1}{n} \sum_i \delta_i^\ell(a) \delta_i^\ell(b).$$

If

$$D_{ab}^\ell = \mathbb{E}[\phi'(Z_a^\ell) \phi'(Z_b^\ell)], \quad Z^\ell \sim N(0, G^{\ell-1}),$$

the fully enumerated expectation peel gives

$$\Pi^3 = D^3, \quad \Pi^2 = D^2 \odot D^3, \quad \Pi^1 = D^1 \odot D^2 \odot D^3. \quad (11.1)$$

The direct Wick contractions form one ladder through the two backward paths. All fixed-three-layer Stein branches are retained until their equality patterns and downstream sums are counted; their orders are explicitly shown to be lower than the ladder in this expectation. This is an executed expectation calculation, not a universal rule that Onsager branches vanish.

The arbitrary-fixed-depth product recursion is a natural one-state extension of this ladder and is recorded in the case-study appendix with the general readout scaling. A complete all-depth diagram induction and the four-path concentration enumeration remain proof obligations. In particular, the statement

$$\text{Var}(\Pi_{n,ab}^\ell) = O(n^{-1})$$

is the expected concentration conclusion under standard fixed-depth moment conditions, but it is not proved by the displayed one-copy calculation.

The subsequent training execution demonstrates a different outcome. A row-distinct family with zero one-copy mean survives after two copies are formed and becomes a leading fresh Gaussian field in the quadratic hidden-Gram coefficient. Thus “off diagonal” and “Onsager” are structural labels, not asymptotic verdicts.

13. Feature-learning jets

13.1 Gradient-flow derivatives

Let the scaled gradient flow be

$$\dot{\theta}_n(\tau) = -P_n \nabla \mathcal{R}_n(\theta_n(\tau)),$$

and define the differential operator

$$\mathcal{V}_n := -(P_n \nabla \mathcal{R}_n) \cdot \nabla.$$

For a smooth scalar observable A_n ,

$$\left. \frac{d^r}{d\tau^r} A_n(\theta_n(\tau)) \right|_{\tau=0} = \mathcal{V}_n^r A_n(\theta_n(0)). \quad (13.1)$$

At fixed r , repeated application of $\mathcal{V}_n$ creates a finite sum of scalar contractions of initialization derivatives of the observable, network, and loss. Those contractions are peeling inputs.

For a hidden activation Gram,

$$G_{n,ab}^\ell(\tau) = \frac{1}{n_\ell} \sum_i h_i^\ell(x_a; \theta_\tau) h_i^\ell(x_b; \theta_\tau),$$

the intended chain is

$$\text{training derivative} \longrightarrow \text{initialization contraction} \longrightarrow \text{peeling state} \longrightarrow \text{explicit Gaussian normal form}. \quad (13.2)$$

13.2 Discrete gradient descent

Fix a number s of gradient-descent steps and a Taylor order p . For sample a , let $\mathcal{L}_a^{(s)}(\eta)$ be the loss after s steps, and define

$$G_{ab}^{\ell,(s)}(\eta) = \frac{1}{n} \left\langle h_a^{\ell,(s)}(\eta), h_b^{\ell,(s)}(\eta) \right\rangle.$$

For fixed $k \leq p$, the coefficients

$$[\eta^k] \mathbb{E}[\mathcal{L}_a^{(s)}(\eta)]$$

and

$$[\eta^k] \mathbb{E}[G_{ab}^{\ell,(s)}(\eta)]$$

are fixed-order initialization derivative observables once their admissibility and scaling are verified. The finite collection through order p is the **order- p feature-learning jet**.

The case study computes the one-step loss coefficient, the first nontrivial hidden-Gram coefficient, and the exact two-Euler-step decomposition in one fixed μP convention. It also shows why a limiting cancellation may not be differentiated as an identity: although the expected finite-width linear Gram coefficient tends to zero, its directional derivative contributes at the next order.

13.3 What a jet does not prove

Fixed-order initialization coefficients do not by themselves establish:

- interchange of width limits with time differentiation;
- concentration of every coefficient;
- a nonzero Taylor radius uniform in width;
- reconstruction of positive-time training from the local jet;
- validity for a number of steps growing like $1/\eta$;
- arbitrary or width-dependent training times.

These are separate analytic questions.

14. Target theorem and its strengthenings

14.1 Conservative target

The first theorem should be formulated for equal-width fully connected MLPs, fixed batch size B , fixed depth L , fixed derivative order, and a fixed number of training steps when updates are included. Fixed positive width ratios and other architectures are planned extensions. Assume enough polynomially controlled activation derivatives for every state actually generated by the peel.

Target Mean-Field Derivative Peeling Theorem. For every admissible scalar derivative observable O_n , there is a deterministic syntactic transformation Peel that recursively eliminates the maximal remaining layer group. Each elimination replaces that group by a finite sum of explicitly specified Gaussian-integral coefficients multiplying admissible lower-group states. The transformation terminates after at most L maximal-layer eliminations and emits a finite Gaussian normal-form DAG. Under the stated regularity, moment, leave-out, and uniform-integrability hypotheses,

$$\lim_{n \rightarrow \infty} \mathbb{E}_{\theta_0}[O_n] = \text{Eval}(\text{Peel}(O_n)).$$

Every integrand, covariance, dimension, coefficient, activation derivative, and equality pattern in the output is constructed from the network, fixed data, and observable syntax.

This is a target statement. Its hypotheses and closure claims have not yet been proved for the entire proposed grammar.

14.2 Finite-state strengthening

The strongest useful extension would show that, for fixed B , derivative order R , training-step count, and observable schema O , the boundary signatures close on state types whose number is independent of width and depth. After common-state sharing, the normal-form DAG would then have at most

$$C_{B,R,O}L$$

nodes, and every Gaussian atom would have dimension at most

$$d_{B,R,O},$$

where both effective constants are independent of width and depth. They may grow combinatorially or exponentially with derivative order and step count; no polynomial bound is currently claimed.

14.3 Possible batch-dimensional factorization

A stronger corollary would reduce each connected Gaussian atom to one B -dimensional layer-preactivation integral, with multiple row components factoring into products. The current examples motivate this in simple states, but no general equality-partition factorization theorem has been proved. The safe target therefore uses $d_{B,R,O}$ rather than asserting dimension B .

15. Claim ladder and relation to Tensor Programs

The proposed contribution must be separated from established prior work.

- Deterministic limits of normalized wide-network averages are broadly established and are not the novelty claim.
- Tensor Programs gives recursive limiting coordinate semantics and moment expectations for broad finite tensor computations, including μP feature-learning limits.
- Recursive first- and higher-derivative kernel analyses also have prior art.
- A mechanical compiler from a specified contracted derivative grammar to a completely eliminated Gaussian normal form is an open target claim.
- Termination after maximal-layer eliminations, depth-independent finite-state closure, and a linear-size shared DAG are progressively stronger open targets.
- Exactly B -dimensional atoms and finite-time dynamics are not established.

15.1 What recursive Tensor Program semantics already provides

For a coordinatewise operation

$$x_i = \phi(y_i^1, \dots, y_i^k),$$

the limiting coordinate variable is recursively represented as

$$Z^x = \phi(Z^{y^1}, \dots, Z^{y^k}).$$

For a normalized average

$$q_n = \frac{1}{n} \sum_i \psi(x_i^1, \dots, x_i^k),$$

the limiting scalar is

$$q_n \longrightarrow \mathbb{E}[\psi(Z^{x^1}, \dots, Z^{x^k})].$$

When a Gaussian matrix W is reused, Tensor Programs decomposes the limiting coordinate schematically as

$$Z^{Wx} = \widehat{Z}^{Wx} + \dot{Z}^{Wx}, \quad (15.1)$$

where the fresh variables generated by the same matrix are jointly Gaussian,

$$\text{Cov}(\widehat{Z}^{Wx}, \widehat{Z}^{Wy}) = \sigma_W^2 \mathbb{E}[Z^x Z^y], \quad (15.2)$$

and a transpose reuse creates an Onsager response of the schematic form

$$\dot{Z}^{Wx} = \sigma_W^2 \sum_a Z^{ya} \mathbb{E} \left[\frac{\partial Z^x}{\partial \widehat{Z}^{W^T y_a}} \right]. \quad (15.3)$$

These are constructive semantics, not an existence-only theorem.

15.2 Intended distinction

Tensor Programs begins with a straight-line tensor computation. A contracted higher derivative must first be differentiated, normalized, compiled into that program language, and checked against the master-theorem hypotheses. Its final answer may retain recursively named nonlinear limit variables and symbolic Onsager responses.

For any particular finite program, one may in principle continue unwinding those recursive variables. If all fresh Gaussian innovations required by the program are collected into a finite vector

$$G \sim \mathcal{N}(0, \Sigma_G),$$

then the recursively defined limit variables can be represented as a deterministic measurable transformation

$$(Z^1, \dots, Z^m) = F(G).$$

Consequently, whenever the required integrability holds,

$$\mathbb{E} \Psi(Z^1, \dots, Z^m) = \int \Psi(F(g)) \mathcal{N}(0, \Sigma_G)(dg).$$

This observation does not by itself provide the proposed peeling normal form. The map F may still contain a recursively nested history of nonlinear and Onsager operations, and mechanically unwinding all symbolic responses can cause a large expression-size blowup. Thus *recursive Gaussian semantics* and an *explicit, normalized Gaussian-integral representation with a controlled finite state* are different output contracts.

Mean-field derivative peeling seeks a narrower source-to-source result. It begins with contracted derivative syntax, executes every Wick and Stein branch, enumerates equality partitions, performs width counting, and emits a normal form containing only deterministic algebra and fully specified Gaussian integrals. On their common domain, the intended validation identity is

$$\text{Peel}(O) = \text{UnrollTP}(\text{CompileDerivative}(O)). \quad (15.4)$$

Equation (15.4) is itself a proposed equivalence theorem, not an assumption. The defensible positioning is therefore a specialized explicit normal-form compiler, not a replacement for Tensor Programs and not the first recursive Gaussian semantics for wide networks.

16. Amendments forced by the executions

The audited examples require the following maintained form of the program.

1. Use layerwise conditional Gaussianity, never an unconditional global Gaussian law across layers.
2. Replace pseudo-independence language by exact Wick–Stein partial matchings. Any limiting independence must follow from a joint law and vanishing mixed covariances.
3. Peel the readout before hidden groups; it fixes parity, scale, and the hidden boundary state.
4. Retain every lower-group factor created by Stein differentiation.
5. Split exact equality patterns before rowwise averaging or independence is used.
6. Count widths globally, including optimizer factors and downstream sums.
7. Do not discard an off-diagonal family merely because its one-copy mean is zero; it may survive in a square as a fresh field.
8. Perform exact conditional integration before deterministic Gram replacement; use leave-finitely-many-out when distinguished rows enter the empirical Gram.
9. Treat an exact subleading expansion separately from a leading mean-field replacement. Gram fluctuations can modify a displayed $1/n$ coefficient.
10. Separate expectation calculations from concentration or fluctuation theorems.
11. Register all channels generated through the same matrix and retain their complete covariance block.
12. Do not differentiate a limiting cancellation as though it were an exact finite-width identity.
13. Permit a finite Gaussian dimension qB or $d_{B,R,O}$ at higher order; a single B -vector is not automatic.
14. Treat $O(L)$ formula size as a theorem conclusion requiring depth-uniform state closure.
15. Fix depth, batch size, derivative order, and step count before the width limit; growing-depth regimes require new uniform estimates.
16. Handle a scalar output separately because it has no neuron-row law-of-large-numbers factor.
17. Allow the activation derivative order generated by the peel to exceed the nominal derivative order of the input observable.
18. Track readout conventions explicitly. A noncentered readout may require backward first-moment/off-diagonal states; a zero readout may make the initialization backward kernel vanish; an $n^{-1/2}$ readout retains a random Gaussian-process output whereas an n^{-1} centered readout tends to zero.
19. Biases require their own Stein branches; their bias–preactivation covariance can be order one.
20. ReLU requires weak derivatives, smoothing, or Gaussian boundary terms; pointwise C^r formulas do not apply directly.

17. Required nonclaims

The current program does not prove:

- convergence of a learning-rate or time Taylor series;
- complete positive-time training dynamics;
- arbitrary or width-dependent training times;
- a uniform-in-depth probability limit;
- elementary closed forms for general activations;
- polynomial complexity in derivative order or training-step count;
- self-averaging of every admissible observable;
- general architectural universality;
- exactly B -dimensional integration without a factorization theorem.

Expectation-level exactness and sample-level determinism are separate claims.

18. Proof obligations

A complete theorem should be divided into the following modules.

1. **Observable grammar and algebraic compilation.** Define the contracted derivative language noncircularly and prove that chain, product, and Faà di Bruno expansion produce finitely many scalar states.
2. **Derivative closure and Stein termination.** Prove that every local elimination remains inside a declared state language and terminates without uncontrolled activation derivative order.
3. **Exact layer elimination.** Establish (7.4) for the chosen regularity class, including weights, biases, rows, and batch coordinates.
4. **Equality-partition completeness.** Enumerate every compatible pattern and prove that no locally subleading branch can recover order through an omitted lower contraction.
5. **Invariant global width degree.** Formalize width valuation under every transition, including optimizer and raw/effective multipliers.
6. **Uniform moments and leave-out estimates.** Bound all retained states and justify removal of finitely many distinguished rows.
7. **Joint conditional CLT.** Prove joint convergence of every registered fresh Gaussian base field and its entire covariance block. Nonlinear states are explicit functions of those bases; they need not themselves be jointly Gaussian.
8. **Covariance replacement and fluctuations.** Justify leading replacement and Price expansions at subleading orders.
9. **Multi-copy concentration.** Show that connected cross-copy diagrams are subleading whenever a deterministic limit is claimed.
10. **Finite-state and depth-uniform closure.** Classify boundary signatures and prove the state-size bound needed for a linear-depth DAG.
11. **Complexity and Gaussian dimension.** State effective dependence on B, R , step count, and observable schema, and determine whether the single- B -vector factorization is valid.
12. **Nonsmooth activations and architectural extensions.** Treat weak derivatives, residual connections, unequal widths, and biases separately.
13. **Tensor Programs equivalence.** Prove (15.4) on the overlap class as a validation theorem.
14. **Generalize the forest factorization.** Determine which parts of the quadratic quotient-forest theorem survive for the full conditional Wick–Stein boundary language, including nonlinear response attachments.

The decisive unresolved step is depth-uniform finite-state closure. Without it, peeling is an explicit layerwise elimination recipe but not yet the claimed $O(L)$ dynamic program.

19. Practical worksheet

19.1 Specify

- What is the raw parameter and what effective tensor enters the forward pass?
- What are the initialization variances and optimizer factors?
- What scalar observable and normalization are used?
- Is the target an expectation, deterministic limit, or fluctuation?
- What width order is requested?
- Which readout, bias, smoothness, and order-of-limits assumptions apply?

19.2 Compile

- Expand only the highest active group.
- Write every current tensor product with explicit indices.
- Record width powers, equality constraints, and the group of every factor.
- Retain all open lower-layer boundary indices.

19.3 Peel

- Condition on lower groups.
- Split exact equality partitions.
- Execute every Wick and Stein branch.
- Differentiate nonlinear factors explicitly.
- Insert exact conditional covariances.
- Carry newly created lower activations downward.
- Count the complete diagram before discarding it.

19.4 Close

- Convert row averages into Gaussian moments.
- Justify empirical-to-deterministic covariance replacement.
- Register fresh Gaussian channels and all cross-covariances.
- Repeat on the next group.
- If the result enters another random product, prove the required multi-copy convergence separately.

19.5 Audit

- Check centered-readout parity.
- Check a linear-activation specialization.
- Check a one-sample or batch-diagonal case.
- Compare width degree with a direct indexed calculation.
- Verify that no limiting zero was differentiated away.
- Verify whether each claim concerns expectation, convergence in probability, L^2 , or a fluctuation law.

20. Terminology, framing, and literature anchors

Recommended vocabulary is:

- **method:** mean-field derivative peeling, shortened to mean-field peeling;
- **target theorem:** Mean-Field Derivative Peeling Theorem;
- **output:** Gaussian normal form;
- **local training data:** feature-learning jet;
- **algorithmic representation:** peeling-state DAG.

The acronym “MFP” is avoided because it is naturally read as mean-field parameterization. A suitable paper title is:

Mean-Field Peeling: Explicit Gaussian Normal Forms for Feature-Learning Jets

If “jet” is undesirable, “Taylor coefficients” is the direct replacement. Other retained editorial alternatives are:

1. **Mean-Field Derivative Peeling: An Explicit Gaussian Calculus for Local Feature Learning;**
2. **Peeling Wide Neural Networks: Explicit Gaussian Recursions for Feature-Learning Derivatives;**
3. **From Derivatives to Gaussian Integrals: Mean-Field Peeling in Feature-Learning Networks;**
4. **Feature-Learning Jets at Infinite Width: A Mean-Field Peeling Calculus.**

“Gaussian Normal-Form Peeling Theorem” and “Layerwise Gaussian Elimination Theorem for Derivative Observables” are descriptive alternatives for the core theorem. “Feature-Learning Jet Representation Theorem” is better reserved for the training corollary.

The literature comparison is an evidence-based positioning assessment, not a certification of novelty. Primary anchors are:

- Greg Yang and Edward J. Hu, [Tensor Programs IV: Feature Learning in Infinite-Width Neural Networks](#), ICML 2021, together with its [supplementary material](#), especially the Master Theorem and Algorithm 1;
- Greg Yang, [Tensor Programs II: Neural Tangent Kernel for Any Architecture](#), 2020;
- Greg Yang, [Tensor Programs III: Neural Matrix Laws](#), 2020;
- Jiaoyang Huang and Horng-Tzer Yau, [Dynamics of Deep Neural Networks and Neural Tangent Hierarchy](#), ICML 2020;
- Ethan Dyer and Guy Gur-Ari, [Asymptotics of Wide Networks from Feynman Diagrams](#), 2019;
- Max Guillen, Philipp Misof, and Jan E. Gerken, [Finite-Width Neural Tangent Kernels from Feynman Diagrams](#), ICML 2026.

21. Current authoritative positioning

Mean-field peeling is a proposed explicit normal-form calculus for fixed-order contracted derivative observables in wide mean-field MLPs. Tensor Programs already supplies constructive recursive semantics for broad finite neural computations, including μP feature-learning limits. The prospective distinction is that peeling begins from derivative syntax and performs all layerwise Gaussian elimination, Stein correction, Wick contraction, equality-pattern enumeration, and width power counting explicitly, with the goal of a finite Gaussian-integral DAG and a finite-state depth recursion.

There are two complementary core laboratories, five bounded parameter- extension campaigns, and one stopped threshold diagnostic. The fixed three-hidden-layer backward and training calculations expose conditional Stein corrections and lower-layer boundary transport. The canonical one-sample quadratic compiler proves an exact decorated-forest specialization and supplies accepted integer coefficients through derivative order eleven. Its campaigns retain the same basic grammar while varying a relative block metric and hidden observable, two- and three-input symmetry channels, a shifted first-hidden activation, and two independent hidden-block metric weights. Four reach exact finite Hankel endpoints downstream; the three-input route stops at lower moments, and the threshold diagnostic contributes no new D13 bound. They provide exact finite-order continuum tests, not a generic compiler theorem. The general nonlinear two-step closure remains formal, its deep-linear specialization is audited, and the finite-state theorem for the full proposed admissible grammar remains open. None of the exact specializations should be mistaken for that general theorem.

Part II. Detailed μ P Training Case Study

Case Study: Mean-Field Peeling for a μ P Network at Initialization, After One Step, and After Two Steps

Status: Maintained worked execution

Date: 2026-08-03

Consolidated: 2026-08-05

Role: A self-contained, low-level execution of the mean-field peeling program on one fixed network and three increasingly difficult observables.

Canonical theory: `CURRENT_RESEARCH_STATE.md`

This case study performs three calculations in one notation:

1. the initialization backward-sensitivity kernels;
2. the output and loss changes after one gradient-descent step through order η , and the hidden-Gram change through order η^2 ;
3. the hidden-Gram changes after two gradient-descent steps, through order η^2 , including the correction caused by updating the gradient itself.

The purpose is not merely to quote the answers. Every calculation is organized in the order required by mean-field peeling:

1. fix the raw/effective parameter and optimizer scalings;
2. expand derivatives and training updates before taking a width limit;
3. scalarize the highest active group;
4. peel the readout first;
5. apply exact conditional Wick–Stein identities to the highest hidden group;
6. retain every lower-group factor created by Stein differentiation;
7. split equality patterns before width counting;
8. replace normalized empirical averages by deterministic Gaussian expectations only after all indices using them have been exposed;
9. repeat downward through the network;
10. audit the result by parity, scaling, and a solvable deep-linear specialization.

The finite-width Taylor and chain-rule identities below are exact. The mean-field limits require the regularity assumptions stated in Section 2.7.

Document map

- Section 1 states the consolidated answers.
- Section 2 fixes one raw-coordinate μ P convention.
- Section 3 derives the Gaussian identities.
- Section 4 executes the backward-kernel peel.
- Section 5 computes the one-step coefficients.
- Section 6 computes the two-step gradient correction.
- Section 7 records amendments to the general program.
- Appendix A records the arbitrary-fixed-depth readout-scaled backward template.
- Appendix B gives the complete five-branch structural formula omitted from the earlier branch-order summary.

Claim-status convention

- **Exact finite-width identity** means no width limit or independence approximation has been used.
- **Mean-field result** means the coefficient follows from the displayed peel under the smoothness, moment, leave-one-out, and concentration assumptions in Section 2.7.
- **Formal mean-field closure** means the finite state and every required Wick–Stein operation have been specified, but a joint convergence theorem for the complete state has not yet been supplied. The general nonlinear two-step recursion has this status; its deep-linear specialization is explicitly audited.

1. Headline results

For the network defined below, let

$$G_{n,ab}^\ell = \frac{1}{n} \sum_i h_i^\ell(a) h_i^\ell(b), \quad \ell = 1, 2, 3,$$

be the hidden activation Gram between batch samples a and b .

At initialization, the executed one-copy peel proves that the expectations of the normalized backward kernels converge to

$$\Pi_{ab}^3 = D_{ab}^3,$$

$$\Pi_{ab}^2 = D_{ab}^2 D_{ab}^3,$$

$$\Pi_{ab}^1 = D_{ab}^1 D_{ab}^2 D_{ab}^3.$$

Conditional on the separate concentration obligation in Section 4.4, the random empirical kernels themselves converge to these limits in L^2 .

For one gradient-descent step, the expected sample loss has the linear expansion

$$\mathbb{E}[(f_a^+ - y_a)^2] = y_a^2 - \frac{4\eta}{B} y_a \sum_b K_{ab} y_b + o(\eta) + o_n(1),$$

where

$$K_{ab} = G_{ab}^3 + G_{ab}^2 D_{ab}^3 + G_{ab}^1 D_{ab}^2 D_{ab}^3 + G_{ab}^0 D_{ab}^1 D_{ab}^2 D_{ab}^3.$$

The expected hidden-Gram linear coefficient vanishes at infinite width:

$$\lim_{n \rightarrow \infty} \frac{d}{d\eta} \mathbb{E}[G_{n,ab}^{\ell,+}] \Big|_{\eta=0} = 0.$$

Its first generic deterministic coefficient is quadratic:

$$\mathbb{E}[G_{n,ab}^{\ell,+}] = G_{ab}^\ell + \eta O(n^{-1}) + \eta^2 C_{ab}^\ell + o(\eta^2) + o_n(1).$$

After two Euler gradient-descent steps,

$$\mathbb{E}[G_{n,ab}^{\ell,(2)}] = G_{ab}^\ell + \eta O(n^{-1}) + \eta^2 (4C_{ab}^\ell + R_{ab}^\ell) + o(\eta^2) + o_n(1),$$

where R^ℓ is the new gradient-correction term. The change produced by the second update alone is

$$\mathbb{E}[G_{n,ab}^{\ell,(2)} - G_{n,ab}^{\ell,(1)}] = \eta O(n^{-1}) + \eta^2 (3C_{ab}^\ell + R_{ab}^\ell) + o(\eta^2) + o_n(1).$$

Neither two-step quantity has a deterministic $O(\eta)$ hidden-Gram term.

2. Fixed network, batch, initialization, and μP update

2.1 Indices

- Batch indices are $a, b, c, d \in \{1, \dots, B\}$.
- First-hidden-layer neuron indices are i, j .
- Second-hidden-layer neuron indices are p, q .
- Third-hidden-layer neuron indices are r, t .
- The input dimension d_0 and batch size B are fixed while $n \rightarrow \infty$.
- Every hidden layer has width n .

2.2 Forward pass

There are three hidden layers and one scalar output. Biases are omitted.

$$\begin{aligned} z_i^1(a) &= \frac{1}{\sqrt{d_0}} \sum_{j=1}^{d_0} W_{ij}^1 x_{aj}, & h_i^1(a) &= \phi(z_i^1(a)), \\ z_p^2(a) &= \frac{1}{\sqrt{n}} \sum_i W_{pi}^2 h_i^1(a), & h_p^2(a) &= \phi(z_p^2(a)), \\ z_r^3(a) &= \frac{1}{\sqrt{n}} \sum_p W_{rp}^3 h_p^2(a), & h_r^3(a) &= \phi(z_r^3(a)), \end{aligned}$$

and

$$f_a = \frac{1}{n} \sum_r v_r h_r^3(a).$$

The raw parameters are independent standard Gaussians:

$$W_{ij}^1, W_{pi}^2, W_{rp}^3, v_r \stackrel{\text{iid}}{\sim} N(0, 1).$$

2.3 Effective parameter variances and learning rates

The effective tensors appearing without explicit normalization would be

$$\frac{W^1}{\sqrt{d_0}}, \quad \frac{W^2}{\sqrt{n}}, \quad \frac{W^3}{\sqrt{n}}, \quad \frac{v}{n}.$$

Their initialization variances and equivalent gradient learning rates are

Effective tensor	Entry variance	Learning rate multiplying its gradient
$W^1/\sqrt{d_0}$	$1/d_0$	$n\eta/d_0$
$W^2/\sqrt{n}$	$1/n$	η
$W^3/\sqrt{n}$	$1/n$	η
v/n	$1/n^2$	η/n

Equivalently, in the raw coordinates used throughout this case study, every block is updated by

$$\theta^+ = \theta - n\eta \nabla_{\theta} \mathcal{J}, \quad \theta \in \{W^1, W^2, W^3, v\}.$$

The fixed d_0 factors are a convention. The displayed width powers are the μP /maximal-update powers.

2.4 Loss and residuals

Use the average squared batch loss

$$\mathcal{J} = \frac{1}{B} \sum_{c=1}^B (f_c - y_c)^2.$$

Write

$$e_c = f_c - y_c.$$

No factor 1/2 is placed in the squared loss, so every gradient contains the factor 2/B.

2.5 Forward activation Grams

Define

$$G_{ab}^0 = \frac{x_a^\top x_b}{d_0},$$

and, at finite width,

$$G_{n,ab}^\ell = \frac{1}{n} \sum_i h_i^\ell(a) h_i^\ell(b), \quad \ell = 1, 2, 3.$$

The deterministic forward recursion is as follows. Let

$$Z^1 \sim N(0, G^0),$$

and define

$$G_{ab}^1 = \mathbb{E}[\phi(Z_a^1) \phi(Z_b^1)].$$

Next let

$$Z^2 \sim N(0, G^1),$$

and define

$$G_{ab}^2 = \mathbb{E}[\phi(Z_a^2) \phi(Z_b^2)].$$

Finally let

$$Z^3 \sim N(0, G^2),$$

and define

$$G_{ab}^3 = \mathbb{E}[\phi(Z_a^3) \phi(Z_b^3)].$$

For each layer define the derivative kernel

$$D_{ab}^\ell = \mathbb{E}[\phi'(Z_a^\ell) \phi'(Z_b^\ell)].$$

Every expectation in these definitions is over one B -dimensional centered Gaussian vector with the displayed covariance.

2.6 Normalized backward variables

Define

$$\delta_i^\ell(a) := n \frac{\partial f_a}{\partial z_i^\ell(a)}.$$

The chain rule gives

$$\delta_r^3(a) = v_r \phi'(z_r^3(a)),$$

$$\delta_p^2(a) = \phi'(z_p^2(a)) \frac{1}{\sqrt{n}} \sum_r W_{rp}^3 \delta_r^3(a),$$

and

$$\delta_i^1(a) = \phi'(z_i^1(a)) \frac{1}{\sqrt{n}} \sum_p W_{pi}^2 \delta_p^2(a).$$

The normalization makes every $\delta_i^\ell(a)$ an $O(1)$ random variable at fixed depth.

2.7 Assumptions and order of limits

The calculations below assume:

1. fixed B, d_0 , and depth;
2. $n \rightarrow \infty$ through equal hidden widths;
3. enough smoothness of ϕ for every displayed Taylor and Stein derivative;
4. polynomial or otherwise controlled growth of the required activation derivatives;
5. uniform moment bounds for every retained contraction state;
6. leave-finitely-many-out control when a singled-out neuron also enters an empirical covariance;
7. uniform integrability when convergence in probability is converted into convergence of expectations;
8. joint conditional multivariate CLTs, including convergence of every mixed covariance state, for the finite collection of forward, backward, tangent, and fresh Gaussian base fields retained by each peel.

The Taylor coefficient is first defined at finite n . The width limit is then taken. Every later display using shorthand such as $o(\eta^2) + o_n(1)$ is to be read in this ordered, coefficientwise sense; no uniform two-parameter remainder estimate is claimed. Claims about a whole finite- η trajectory require additional uniform remainder estimates and are not implied merely by the coefficient calculations.

For ReLU, ordinary pointwise C^2 Taylor formulas are unavailable. A weak-derivative or Gaussian-boundary calculation is required.

3. The two exact Gaussian identities used in every peel

Let $w = (w_1, \dots, w_n)$ have independent $N(0, 1)$ coordinates and let

$$Z(a) = \frac{1}{\sqrt{n}} \sum_j w_j h_j(a).$$

Conditional on the lower-layer numbers $h_j(a)$, the vector (w, Z) is jointly Gaussian.

For a smooth function F of the fixed-batch vector Z , one-weight Stein gives

$$\mathbb{E}[w_p F(Z)] = \frac{1}{\sqrt{n}} \sum_a h_p(a) \mathbb{E}[\partial_a F(Z)]. \quad (3.1)$$

Applying Gaussian integration by parts twice gives

$$\mathbb{E}[w_p w_q F(Z)] = \delta_{pq} \mathbb{E}[F(Z)] + \frac{1}{n} \sum_{a,b} h_p(a) h_q(b) \mathbb{E}[\partial_a \partial_b F(Z)]. \quad (3.2)$$

The first term of (3.2) is the weight-weight Wick branch. The second is the Stein/Onsager branch. No independence between w and Z has been asserted.

For a top-row sum, equality patterns must be separated. For example, $p = q$ and $p \neq q$ have different numbers of free indices and different Gaussian contractions. A branch is discarded only after its complete index multiplicity and every lower-layer sum created by Stein differentiation have been counted.

4. Execution I: backward sensitivity kernels

Define

$$\Pi_{n,ab}^\ell = \frac{1}{n} \sum_i \delta_i^\ell(a) \delta_i^\ell(b).$$

We peel from the readout downward.

4.1 Readout and group 3

At the third hidden layer,

$$\Pi_{n,ab}^3 = \frac{1}{n} \sum_r v_r^2 \phi'(z_r^3(a)) \phi'(z_r^3(b)).$$

The readout weights are independent of all hidden preactivations, so

$$\mathbb{E}[v_r^2] = 1.$$

Conditional on the second hidden layer, the rows z_r^3 are iid Gaussian with covariance G_n^2 . A weighted law of large numbers followed by $G_n^2 \rightarrow G^2$ gives

$$\boxed{\Pi_{n,ab}^3 \longrightarrow D_{ab}^3.}$$

4.2 Full scalarization of the group-3 peel for Π^2

Substitute the recursion for δ^2 :

$$\begin{aligned} \Pi_{n,ab}^2 &= \frac{1}{n^2} \sum_{p,r,t} \phi'(z_p^2(a)) \phi'(z_p^2(b)) W_{rp}^3 W_{tp}^3 \\ &\quad \times v_r v_t \phi'(z_r^3(a)) \phi'(z_t^3(b)). \end{aligned} \quad (4.1)$$

The readout is the highest group. Peel it first:

$$\mathbb{E}_v[v_r v_t] = \delta_{rt}.$$

Thus

$$\begin{aligned} \mathbb{E}_v[\Pi_{n,ab}^2] &= \frac{1}{n^2} \sum_{p,r} \phi'(z_p^2(a)) \phi'(z_p^2(b)) (W_{rp}^3)^2 \\ &\quad \times \phi'(z_r^3(a)) \phi'(z_r^3(b)). \end{aligned} \quad (4.2)$$

The factor $(W_{rp}^3)^2$ is not independent of z_r^3 . Apply (3.2) with $p = q$ and

$$F(Z) = \phi'(Z_a) \phi'(Z_b).$$

Because

$$\frac{\partial z_r^3(c)}{\partial W_{rp}^3} = \frac{h_p^2(c)}{\sqrt{n}},$$

the exact second derivative is

$$\begin{aligned} &\frac{\partial^2}{\partial (W_{rp}^3)^2} [\phi'(z_r^3(a)) \phi'(z_r^3(b))] \\ &= \frac{1}{n} \left[h_p^2(a)^2 \phi'''(z_r^3(a)) \phi'(z_r^3(b)) \right. \\ &\quad + 2h_p^2(a) h_p^2(b) \phi''(z_r^3(a)) \phi''(z_r^3(b)) \\ &\quad \left. + h_p^2(b)^2 \phi'(z_r^3(a)) \phi'''(z_r^3(b)) \right]. \end{aligned} \quad (4.3)$$

Let

$$\mathcal{E}_{n,ab}^3(c, d) = \mathbb{E}_{Z \sim N(0, G_n^2)} [\partial_c \partial_d \{\phi'(Z_a) \phi'(Z_b)\}].$$

After applying (3.2) and summing the group-3-only index r , the exact conditional expression is

$$\begin{aligned} &\mathbb{E}_{v, W^3}[\Pi_{n,ab}^2 \mid W^1, W^2] \\ &= \frac{1}{n} \sum_p \phi'(z_p^2(a)) \phi'(z_p^2(b)) \left[D_{n,ab}^3 + \frac{1}{n} \sum_{c,d} h_p^2(c) h_p^2(d) \mathcal{E}_{n,ab}^3(c, d) \right], \end{aligned} \quad (4.3a)$$

where

$$D_{n,ab}^3 = \mathbb{E}_{Z \sim N(0, G_n^2)} [\phi'(Z_a) \phi'(Z_b)].$$

Only now is the empirical covariance replaced by its deterministic limit. The first branch of (4.3a) converges to $D_{ab}^2 D_{ab}^3$. The second branch has the form $n^{-2} \sum_p$ of bounded-moment row terms and is $O(n^{-1})$. Therefore

$$\boxed{\mathbb{E}[\Pi_{n,ab}^2] \longrightarrow D_{ab}^2 D_{ab}^3.}$$

4.3 Full path scalarization for Π^1

Expanding both backward paths gives

$$\begin{aligned} \Pi_{n,ab}^1 &= \frac{1}{n^3} \sum_{i,p,q,r,t} v_r v_t W_{rp}^3 W_{iq}^3 W_{pi}^2 W_{qt}^2 \\ &\quad \times \phi'(z_i^1(a)) \phi'(z_i^1(b)) \\ &\quad \times \phi'(z_p^2(a)) \phi'(z_q^2(b)) \\ &\quad \times \phi'(z_r^3(a)) \phi'(z_t^3(b)). \end{aligned} \quad (4.4)$$

The index ownership is essential:

- after the readout peel, r is group-3-only;
- p, q occur in groups 3 and 2 and cannot be summed out during the group-3 peel;
- i occurs in groups 2 and 1 and cannot be summed out during the group-2 peel.

4.3.1 Peel the readout

Using

$$\mathbb{E}[v_r v_t] = \delta_{rt},$$

we obtain

$$\begin{aligned} \mathbb{E}_v[\Pi_{n,ab}^1] &= \frac{1}{n^3} \sum_{i,p,q,r} W_{rp}^3 W_{rq}^3 W_{pi}^2 W_{qi}^2 \\ &\quad \times \phi'(z_i^1(a)) \phi'(z_i^1(b)) \\ &\quad \times \phi'(z_p^2(a)) \phi'(z_q^2(b)) \\ &\quad \times \phi'(z_r^3(a)) \phi'(z_r^3(b)). \end{aligned} \quad (4.5)$$

4.3.2 Peel group 3

Condition on groups 1 and 2. Applying (3.2) to row r gives

$$\begin{aligned} \mathbb{E}_{W^3} [W_{rp}^3 W_{rq}^3 \phi'(z_r^3(a)) \phi'(z_r^3(b))] \\ = \delta_{pq} D_{n,ab}^3 + \frac{1}{n} \sum_{c,d} h_p^2(c) h_q^2(d) \mathcal{E}_{n,ab}^3(c, d), \end{aligned} \quad (4.6)$$

where

$$D_{n,ab}^3 = \mathbb{E}_{Z \sim N(0, G_n^2)} [\phi'(Z_a) \phi'(Z_b)],$$

and

$$\mathcal{E}_{n,ab}^3(c, d) = \mathbb{E}_{Z \sim N(0, G_n^2)} [\partial_c \partial_d \{\phi'(Z_a) \phi'(Z_b)\}].$$

The Stein-created factors $h_p^2(c) h_q^2(d)$ belong to group 2. They remain inside the lower boundary state.

The first branch of (4.6) enforces $p = q$. The second leaves both patterns $p = q$ and $p \neq q$ to be counted at group 2.

Before applying a group-2 Stein identity, remove the dependence of the coefficients in (4.6) on the distinguished group-2 rows. Using a leave-two-out version of G_n^2 and then its deterministic limit gives

$$D_{n,ab}^3 = D_{ab}^3 + o_{L^1}(1),$$

$$\mathcal{E}_{n,ab}^3(c, d) = \mathcal{E}_{ab}^3(c, d) + o_{L^1}(1), \quad (4.6a)$$

where

$$\mathcal{E}_{ab}^3(c, d) = \mathbb{E}_{Z \sim N(0, G^2)} [\partial_c \partial_d \{\phi'(Z_a) \phi'(Z_b)\}].$$

The $o(1)$ error is carried through the remaining normalized sums by uniform integrability. Equivalently, one could retain derivatives of D_n^3 and $\mathcal{E}_n^3$ with respect to the distinguished rows; the leave-two-out replacement is the cleaner leading-order implementation used here.

4.3.3 Peel group 2: Wick branch

The group-3 Wick branch becomes

$$\frac{D_{ab}^3}{n^2} \sum_{i,p} (W_{pi}^2)^2 \phi'(z_i^1(a)) \phi'(z_i^1(b)) \phi'(z_p^2(a)) \phi'(z_p^2(b)). \quad (4.7)$$

Applying (3.2) to $(W_{pi}^2)^2$ gives a leading Wick term

$$\frac{D_{ab}^3 D_{n,ab}^2}{n} \sum_i \phi'(z_i^1(a)) \phi'(z_i^1(b)), \quad (4.8)$$

up to the already recorded $o(1)$ covariance-replacement error.

More explicitly, if

$$\mathcal{E}_{n,ab}^2(c, d) = \mathbb{E}_{Z \sim N(0, G_n^1)} [\partial_c \partial_d \{\phi'(Z_a) \phi'(Z_b)\}],$$

then

$$\begin{aligned} \mathbb{E}_{W^2} [(W_{pi}^2)^2 \phi'(z_p^2(a)) \phi'(z_p^2(b))] \\ = D_{n,ab}^2 + \frac{1}{n} \sum_{c,d} h_i^1(c) h_i^1(d) \mathcal{E}_{n,ab}^2(c, d). \end{aligned} \quad (4.8a)$$

After the p - and i -sums, the second line contributes $O(n^{-1})$ and leaves only group-1 activations.

4.3.4 Peel group 2: branch created by group-3 Stein differentiation

After the group-3-only index r has been summed, the full branch created by the second term of (4.6) is

$$\begin{aligned} \mathcal{S}_n &= \frac{1}{n^3} \sum_{i,p,q} W_{pi}^2 W_{qi}^2 \phi'(z_i^1(a)) \phi'(z_i^1(b)) \\ &\quad \times \sum_{c,d} \mathcal{C}_{ab}^3(c,d) \phi'(z_p^2(a)) \phi(z_p^2(c)) \\ &\quad \times \phi'(z_q^2(b)) \phi(z_q^2(d)). \end{aligned} \quad (4.9)$$

For $p = q$, define

$$P_{cd}(Z) = \phi'(Z_a) \phi'(Z_b) \phi(Z_c) \phi(Z_d).$$

Conditional on group 1, let $Z \sim N(0, G_n^1)$. The exact group-2 row peel of the row-dependent factor is

$$\begin{aligned} \mathbb{E}[(W_{pi}^2)^2 P_{cd}(z_p^2)] \\ = \mathbb{E}[P_{cd}(Z)] + \frac{1}{n} \sum_{u,w} h_i^1(u) h_i^1(w) \mathbb{E}[\partial_u \partial_w P_{cd}(Z)]. \end{aligned} \quad (4.10)$$

The diagonal family contains only n^2 choices for (i, p) . With the n^{-3} normalization in (4.9), its Wick and Stein terms are respectively $O(n^{-1})$ and $O(n^{-2})$.

For $p \neq q$, the two group-2 rows are conditionally independent. Each single weight must be removed using the one-weight identity (3.1). The two Stein covariances contribute n^{-1} together. There are $n^3(1 + o(1))$ choices for (i, p, q) , but the original normalization is n^{-3} , so this branch is again $O(n^{-1})$.

In full notation, put

$$F_{a,c}(Z) = \phi'(Z_a) \phi(Z_c), \quad F_{b,d}(Z) = \phi'(Z_b) \phi(Z_d).$$

For $p \neq q$,

$$\begin{aligned} \mathbb{E}[W_{pi}^2 F_{a,c}(z_p^2)] &= \frac{1}{\sqrt{n}} \sum_u h_i^1(u) \mathbb{E}[\partial_u F_{a,c}(Z)], \\ \mathbb{E}[W_{qi}^2 F_{b,d}(z_q^2)] &= \frac{1}{\sqrt{n}} \sum_w h_i^1(w) \mathbb{E}[\partial_w F_{b,d}(Z)]. \end{aligned} \quad (4.11)$$

Their product supplies the stated n^{-1} Stein cost. Thus the five branch orders—leading ladder, group-2 Stein correction, group-3-Stein diagonal Wick, group-3-Stein diagonal Stein, and group-3-Stein off-diagonal—are

$$1, \quad n^{-1}, \quad n^{-1}, \quad n^{-2}, \quad n^{-1}. \quad (4.12)$$

Thus every branch created by the group-3 Stein term vanishes at leading order, but only after the $p = q$ and $p \neq q$ families have been separated and counted.

Before the final group-1 law of large numbers, use a leave-one-out covariance to replace

$$D_{n,ab}^2 = D_{ab}^2 + o_{L^1}(1), \quad \mathcal{C}_{n,ab}^2(c,d) = \mathcal{C}_{ab}^2(c,d) + o_{L^1}(1). \quad (4.12a)$$

This prevents the distinguished group-1 row from being hidden inside a coefficient that is incorrectly treated as constant.

4.3.5 Peel group 1

After group 2 is removed, the leading remaining expression is (4.8). The index i now belongs only to group 1, so the law of large numbers gives

$$\frac{1}{n} \sum_i \phi'(z_i^1(a)) \phi'(z_i^1(b)) \longrightarrow D_{ab}^1.$$

Therefore

$$\mathbb{E}[\Pi_{n,ab}^1] \longrightarrow D_{ab}^1 D_{ab}^2 D_{ab}^3.$$

4.4 Result and concentration obligation

The limiting backward recursion is

$$\Pi^3 = D^3, \quad \Pi^2 = D^2 \odot \Pi^3, \quad \Pi^1 = D^1 \odot \Pi^2.$$

Here $\odot$ denotes entrywise multiplication.

The calculation above establishes the expectation limits. To obtain convergence of the random empirical kernel itself, one must peel the square. Four backward paths appear. The expected mechanism is that the disconnected pair of ladders reproduces the square of the expectation, while every connected equality pattern loses at least one free neuron index. A sufficient estimate is

$$\text{Var}(\Pi_{n,ab}^\ell) = O(n^{-1}).$$

The complete four-path Wick–Stein enumeration is not reproduced in this case study. Accordingly, the variance estimate is recorded as an additional concentration hypothesis/proof obligation under the fixed-depth moment assumptions. When it holds,

$$\Pi_{n,ab}^\ell \rightarrow \Pi_{ab}^\ell$$

in L^2 . Every later replacement of a product involving Π_n^ℓ invokes this concentration estimate together with uniform integrability; it does not follow from the one-copy expectation calculation alone.

4.5 Exact conditional integration versus mean-field replacement

At group 3 the Gaussian covariance used in (4.3a) and (4.6) is initially

$$G_{n,cd}^2 = \frac{1}{n} \sum_p h_p^2(c) h_p^2(d),$$

not the deterministic matrix G^2 . Likewise, the group-2 row peel initially uses

$$G_{n,cd}^1 = \frac{1}{n} \sum_i h_i^1(c) h_i^1(d).$$

The correct order is:

1. condition on the lower groups and apply the exact Gaussian integration-by-parts identity with $G_n^{\ell-1}$;
2. retain every lower-group factor created by differentiation;
3. expose the equality patterns of the distinguished rows;
4. then replace $G_n^{\ell-1}$ by $G^{\ell-1}$.

When a distinguished row p or pair (p, q) also occurs inside G_n^2 , a quantitative proof replaces G_n^2 temporarily by its leave-one-out or leave-two-out version. Removing finitely many rows changes the covariance by $O(n^{-1})$; the remaining covariance is independent of the distinguished rows and converges to G^2 . Uniform integrability then permits passage to expectations.

This procedure proves the leading $o(1)$ limit. It does not, without a sharper fluctuation expansion, identify an exact finite-width $1/n$ bias.

5. Execution II: one gradient-descent step

The output and sample losses are computed through linear order in η . The hidden Grams are computed through quadratic order, because their deterministic linear coefficient vanishes. All Taylor coefficients are formed at finite width before taking $n \rightarrow \infty$.

5.1 Exact parameter and preactivation velocities

Put a dot over a quantity to mean its derivative with respect to the one-step learning-rate parameter at $\eta = 0$. From

$$\theta^+ = \theta - n\eta \nabla_{\theta} \mathcal{F}$$

and the definition of δ , the exact raw-parameter velocities are

$$\dot{v}_r = -\frac{2}{B} \sum_c e_c h_r^3(c), \quad (5.1)$$

$$\dot{W}_{rp}^3 = -\frac{2}{B\sqrt{n}} \sum_c e_c \delta_r^3(c) h_p^2(c), \quad (5.2)$$

$$\dot{W}_{pi}^2 = -\frac{2}{B\sqrt{n}} \sum_c e_c \delta_p^2(c) h_i^1(c), \quad (5.3)$$

and

$$\dot{W}_{ij}^1 = -\frac{2}{B\sqrt{d_0}} \sum_c e_c \delta_i^1(c) x_{cj}. \quad (5.4)$$

Define the first preactivation velocities

$$u_i^\ell(a) := \left. \frac{dz_i^{\ell,+}(a)}{d\eta} \right|_{\eta=0}.$$

Substitution of (5.2)–(5.4) into the forward equations gives, still exactly at finite width,

$$u_i^1(a) = -\frac{2}{B} \sum_c e_c G_{ac}^0 \delta_i^1(c), \quad (5.5)$$

$$\begin{aligned} u_p^2(a) = & -\frac{2}{B} \sum_c e_c G_{n,ac}^1 \delta_p^2(c) \\ & + \frac{1}{\sqrt{n}} \sum_i W_{pi}^2 \phi'(z_i^1(a)) u_i^1(a), \end{aligned} \quad (5.6)$$

and

$$\begin{aligned}
u_r^3(a) = & -\frac{2}{B} \sum_c e_c G_{n,ac}^2 \delta_r^3(c) \\
& + \frac{1}{\sqrt{n}} \sum_p W_{rp}^3 \phi'(z_p^2(a)) u_p^2(a).
\end{aligned} \tag{5.7}$$

These equations already exhibit the two branches that peeling must track: the direct change of the current matrix and the propagated change of the preceding representation.

5.2 Exact tangent kernel and the linear loss coefficient

Define the optimizer-metric tangent kernel

$$K_{n,ab} = n \sum_{k=1}^3 \langle \nabla_{W^k} f_a, \nabla_{W^k} f_b \rangle + n \langle \nabla_v f_a, \nabla_v f_b \rangle. \tag{5.8}$$

The finite-width output velocity is therefore

$$\dot{f}_a = -\frac{2}{B} \sum_c K_{n,ac} e_c. \tag{5.9}$$

Every parameter gradient is rank one. For example,

$$\frac{\partial f_a}{\partial W_{rp}^3} = \frac{\delta_r^3(a) h_p^2(a)}{n \sqrt{n}},$$

so

$$\begin{aligned}
n \sum_{r,p} \frac{\partial f_a}{\partial W_{rp}^3} \frac{\partial f_b}{\partial W_{rp}^3} &= \left(\frac{1}{n} \sum_r \delta_r^3(a) \delta_r^3(b) \right) \left(\frac{1}{n} \sum_p h_p^2(a) h_p^2(b) \right) \\
&= \Pi_{n,ab}^3 G_{n,ab}^2.
\end{aligned}$$

Doing the same for every block gives the exact factorization

$$K_{n,ab} = G_{n,ab}^3 + G_{n,ab}^2 \Pi_{n,ab}^3 + G_{n,ab}^1 \Pi_{n,ab}^2 + G_{n,ab}^0 \Pi_{n,ab}^1. \tag{5.10}$$

The backward expectation calculation in Section 4, together with the joint concentration obligation stated in Section 4.4, gives

$$\begin{aligned}
K_{ab} = & G_{ab}^3 + G_{ab}^2 D_{ab}^3 + G_{ab}^1 D_{ab}^2 D_{ab}^3 \\
& + G_{ab}^0 D_{ab}^1 D_{ab}^2 D_{ab}^3.
\end{aligned} \tag{5.11}$$

Because $f_a \rightarrow 0$, the residual satisfies $e_a \rightarrow -y_a$. Weighted convergence, rather than a finite-width independence claim, yields

$$\lim_{n \rightarrow \infty} \mathbb{E}[\dot{f}_a] = \frac{2}{B} \sum_c K_{ac} y_c. \tag{5.12}$$

For the individual loss $\ell_a = e_a^2$,

$$\dot{\ell}_a = 2e_a \dot{f}_a = -\frac{4}{B} e_a \sum_c K_{n,ac} e_c.$$

Consequently,

$$\mathbb{E}[\ell_a^+] = y_a^2 - \frac{4\eta}{B} y_a \sum_c K_{ac} y_c + o(\eta) + o_n(1). \tag{5.13}$$

For the average loss this becomes

$$\mathbb{E}[\mathcal{J}^+] = \frac{1}{B} \sum_a y_a^2 - \frac{4\eta}{B^2} y^\top K y + o(\eta) + o_n(1). \tag{5.14}$$

The batch-average coefficient is nonpositive because K is positive semidefinite. A particular sample loss need not decrease.

5.3 Exact hidden-Gram linear coefficient

Define the activation velocity

$$g_i^\ell(a) := \left. \frac{dh_i^{\ell,+}(a)}{d\eta} \right|_0 = \phi'(z_i^\ell(a)) u_i^\ell(a). \tag{5.15}$$

The coefficient of η in the hidden Gram is exactly

$$L_{n,ab}^\ell := \left. \frac{dG_{n,ab}^{\ell,+}}{d\eta} \right|_0 = \frac{1}{n} \sum_i \left[g_i^\ell(a) h_i^\ell(b) + h_i^\ell(a) g_i^\ell(b) \right]. \quad (5.16)$$

Equivalently, before any width limit,

$$L_{n,ab}^\ell = -\frac{2n}{B} \sum_c e_c \sum_{k \leq \ell} \left\langle \nabla_{W^k} G_{n,ab}^\ell, \nabla_{W^k} f_c \right\rangle. \quad (5.17)$$

The readout must be peeled before any hidden group. For a coordinate $z_j^k(c)$ with $k \leq 3$,

$$\delta_j^k(c) = \sum_r v_r \phi'(z_r^3(c)) \frac{\partial z_r^3(c)}{\partial z_j^k(c)}$$

is linear in v . Hence the label branch is odd and vanishes:

$$\mathbb{E}_v[-y_c \delta_j^k(c)] = 0. \quad (5.18)$$

The output branch contains two readout factors. Its contraction is

$$\begin{aligned} \mathbb{E}_v[f_c \delta_j^k(c)] &= \frac{1}{n} \sum_{r,t} \mathbb{E}[v_t v_r] h_t^3(c) \phi'(z_r^3(c)) \frac{\partial z_r^3(c)}{\partial z_j^k(c)} \\ &= \frac{1}{n} \sum_r h_r^3(c) \phi'(z_r^3(c)) \frac{\partial z_r^3(c)}{\partial z_j^k(c)} \\ &= \frac{1}{2} \frac{\partial G_{n,cc}^3}{\partial z_j^k(c)}. \end{aligned} \quad (5.19)$$

Substituting this exact top-group peel into (5.17) gives

$$\mathbb{E}_v[L_{n,ab}^\ell \mid W^1, W^2, W^3] = -\frac{1}{B} \sum_c \sum_{k \leq \ell} \left\langle \nabla_{W^k} G_{n,ab}^\ell, \nabla_{W^k} G_{n,cc}^3 \right\rangle. \quad (5.20)$$

This formula makes the remaining width loss visible. For example, the direct $k = \ell = 3$ term is

$$\begin{aligned} &-\frac{2}{Bn} \sum_c \left[G_{n,ac}^2 \frac{1}{n} \sum_r h_r^3(b) \phi'(z_r^3(a)) h_r^3(c) \phi'(z_r^3(c)) \right. \\ &\quad \left. + G_{n,bc}^2 \frac{1}{n} \sum_r h_r^3(a) \phi'(z_r^3(b)) h_r^3(c) \phi'(z_r^3(c)) \right]. \end{aligned} \quad (5.21)$$

The two inner averages have finite Gaussian limits, while the outer $1/n$ remains. For $k < \ell$, expanding the two derivatives in (5.20) creates two forward paths. Peeling from group 3 downward gives either a Wick pairing or a Stein attachment at each shared matrix. Those operations change the limiting Gaussian moment but do not restore the one free neuron index already lost in (5.20). Thus every such path family is also $O(n^{-1})$.

Therefore

$$\mathbb{E}[L_{n,ab}^\ell] = O(n^{-1}),$$

and

$$\lim_{n \rightarrow \infty} [\eta] \mathbb{E}[G_{n,ab}^{\ell,+}] = 0. \quad (5.22)$$

The cancellation is an annealed statement caused by the centered readout. It does not say that an individual finite-width network has zero first-order feature motion.

5.4 Exact quadratic hidden-Gram expansion

At finite width write

$$z_i^{\ell,+}(a) = z_i^\ell(a) + \eta u_i^\ell(a) + \eta^2 q_i^\ell(a) + o(\eta^2), \quad (5.23)$$

and

$$h_i^{\ell,+}(a) = h_i^\ell(a) + \eta g_i^\ell(a) + \eta^2 s_i^\ell(a) + o(\eta^2). \quad (5.24)$$

Taylor's formula gives

$$g_i^\ell(a) = \phi'(z_i^\ell(a)) u_i^\ell(a), \quad (5.25)$$

$$s_i^\ell(a) = \phi'(z_i^\ell(a))q_i^\ell(a) + \frac{1}{2}\phi''(z_i^\ell(a))u_i^\ell(a)^2. \quad (5.26)$$

The first preactivation is linear in W^1 , so

$$q^1 = 0.$$

For $\ell = 2, 3$, expanding the product of the updated matrix and the updated preceding activation gives

$$q_i^\ell(a) = \frac{1}{\sqrt{n}} \sum_j [W_{ij}^\ell s_j^{\ell-1}(a) + \dot{W}_{ij}^\ell g_j^{\ell-1}(a)]. \quad (5.27)$$

Therefore the exact coefficient of η^2 in the hidden Gram is

$$C_{n,ab}^\ell = \frac{1}{n} \sum_i [g_i^\ell(a)g_i^\ell(b) + s_i^\ell(a)h_i^\ell(b) + h_i^\ell(a)s_i^\ell(b)]. \quad (5.28)$$

5.5 Peel the residual and readout factors in the quadratic coefficient

Every first velocity is linear in the residuals. Its leading quadratic products therefore contain

$$e_c e_d = y_c y_d - y_c f_d - y_d f_c + f_c f_d. \quad (5.29)$$

Each normalized backward signal δ is odd under $v \mapsto -v$. A quadratic velocity term has two such signals and is even. The two mixed terms in (5.29) add one further readout factor and are odd, so their conditional readout expectation is zero. The $f_c f_d$ branch is even but is $O(n^{-1})$ in every fixed moment because $f_c = O_{L^p}(n^{-1/2})$. Thus only the label–label term survives at leading order.

Set

$$\lambda_c := \frac{2y_c}{B}. \quad (5.30)$$

From this point through the end of the one-step quadratic calculation, all velocities mean their leading label-driven branch.

5.6 Separate the local activation gate from the incoming backward signal

At finite width define the incoming backward signals by

$$\begin{aligned} \xi_r^3(a) &= v_r, \\ \xi_p^2(a) &= \frac{1}{\sqrt{n}} \sum_r W_{rp}^3 \delta_r^3(a), \\ \xi_i^1(a) &= \frac{1}{\sqrt{n}} \sum_p W_{pi}^2 \delta_p^2(a). \end{aligned} \quad (5.31a)$$

Then the exact identity is

$$\delta_i^\ell(a) = \phi'(z_i^\ell(a))\xi_i^\ell(a).$$

For a limiting representative neuron, write

$$\delta^\ell(a) = \phi'(Z_a^\ell)\xi^\ell(a). \quad (5.31)$$

The multichannel extension of the Section 4 peel gives, under the joint-CLT and leave-one-out assumptions in Section 2.7, Gaussian base fields satisfying

$$Z^\ell \perp \xi^\ell,$$

with

$$\mathbb{E}[\xi^\ell(a)\xi^\ell(b)] = P_{ab}^\ell,$$

where

$$P_{ab}^3 = 1, \quad P_{ab}^2 = D_{ab}^3, \quad P_{ab}^1 = D_{ab}^2 D_{ab}^3. \quad (5.32)$$

Accordingly,

$$\Pi_{ab}^\ell = D_{ab}^\ell P_{ab}^\ell.$$

The displayed independence is an additional leading joint-state conclusion, not a consequence of the backward second moment alone. It follows by conditioning on the forward network, peeling the mixed moments and conditional covariance of ξ^ℓ , removing finitely many distinguished rows, and applying the conditional multivariate CLT. It is not a finite-width claim that a backward signal is independent of the weights and preactivations that define it.

Define three partial tangent factors, entrywise in the pair (a, b) :

$$T^0 = G^0,$$

$$T^1 = G^1 + D^1 \odot T^0,$$

$$T^2 = G^2 + D^2 \odot T^1. \quad (5.33)$$

Thus $T^{\ell-1}$ records the parameters at and below layer ℓ that can move z^ℓ .

5.7 Execute the first-velocity peel

At layer 1, (5.5) immediately becomes

$$u_i^1(a) = \sum_c \lambda_c T_{ac}^0 \delta_i^1(c). \quad (5.34)$$

At layer 2, substitute (5.34) and the explicit transpose channel

$$\xi_i^1(c) = \frac{1}{\sqrt{n}} \sum_{p'} W_{p'i}^2 \delta_{p'}^2(c)$$

into the propagated term of (5.6). This exposes all indices:

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sum_i W_{pi}^2 \phi'(z_i^1(a)) u_i^1(a) \\ &= \frac{1}{n} \sum_{i,p'} W_{pi}^2 W_{p'i}^2 \phi'(z_i^1(a)) \sum_c \lambda_c G_{ac}^0 \phi'(z_i^1(c)) \delta_{p'}^2(c). \end{aligned} \quad (5.35)$$

The equality patterns $p' = p$ and $p' \neq p$ must be treated separately.

For $p' = p$, the Wick branch of the two-weight identity forces the two paths to use the same target row. After the group-1 average is exposed, it gives

$$\sum_c \lambda_c G_{ac}^0 D_{ac}^1 \delta_p^2(c). \quad (5.36)$$

Adding the direct matrix-update term in (5.6),

$$\sum_c \lambda_c G_{ac}^1 \delta_p^2(c),$$

produces

$$\sum_c \lambda_c T_{ac}^1 \delta_p^2(c). \quad (5.37)$$

For $p' \neq p$, an individual term has zero centered conditional mean, but the whole family cannot be discarded: it contains $n^2(1 + o(1))$ pairs (i, p') . Scalarizing two copies of (5.35), pairing the row-distinct weights, and then applying the group-1 law of large numbers leaves a centered Gaussian field. Call it Γ^2 . Its covariance is exactly the preceding activation-velocity covariance,

$$\mathbb{E}[\Gamma^2(a) \Gamma^2(b)] = \mathbb{E}[g^1(a) g^1(b)]. \quad (5.38)$$

The same diagonal-response/off-diagonal-fluctuation split at layer 3 yields the following equality in the limiting representative-neuron law:

$$u^\ell(a) = \Gamma^\ell(a) + \sum_c \lambda_c T_{ac}^{\ell-1} \delta^\ell(c), \quad \ell = 1, 2, 3, \quad (5.39)$$

where $\Gamma^1 = 0$. For $\ell = 2, 3$, the fresh field is centered Gaussian and

$$\mathbb{E}[\Gamma^\ell(a) \Gamma^\ell(b)] = V_{ab}^{\ell-1}, \quad (5.40)$$

where

$$V_{ab}^\ell := \mathbb{E}[g^\ell(a) g^\ell(b)]. \quad (5.41)$$

The remaining cross-covariance with the current preactivation is

$$\mathbb{E}[\Gamma^\ell(a) Z_b^\ell] = \mathbb{E}[g^{\ell-1}(a) h^{\ell-1}(b)].$$

The right-hand side is odd in the centered readout and is zero at this order. The recursion below additionally uses the corresponding mixed two-copy conclusion

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\left(\frac{1}{\sqrt{n}} \sum_j W_{ij}^\ell g_j^{\ell-1}(a) - \text{its complete Stein response} \right) \xi_i^\ell(b) \right] = 0. \quad (5.40a)$$

Its disconnected branch vanishes by readout parity; every connected branch must reconnect the two copies through a shared row or a Stein attachment and loses one width power. A complete joint-state theorem must include this mixed contraction explicitly. Under that joint-CLT obligation, zero cross-covariance of the Gaussian base fields makes Γ^ℓ independent of (Z^ℓ, ξ^ℓ) at this one-step order.

5.8 Recursion for the velocity covariance

Insert (5.31) and (5.39) into

$$g^\ell(a) = \phi'(Z_a^\ell)u^\ell(a).$$

The fresh-field/response cross terms vanish. The fresh-field square gives $D_{ab}^\ell V_{ab}^{\ell-1}$. The response square gives one backward covariance P_{cd}^ℓ and a four-gate Gaussian moment. Therefore

$$\boxed{V_{ab}^\ell = D_{ab}^\ell V_{ab}^{\ell-1} + \sum_{c,d} \lambda_c \lambda_d T_{ac}^{\ell-1} T_{bd}^{\ell-1} P_{cd}^\ell \times \mathbb{E}[\phi'(Z_a^\ell) \phi'(Z_b^\ell) \phi'(Z_c^\ell) \phi'(Z_d^\ell)],} \quad (5.42)$$

with $V^0 = 0$ and $Z^\ell \sim N(0, G^{\ell-1})$.

The first line of (5.42) is the off-diagonal Wick/CLT branch just derived from $p' \neq p$. Replacing the empirical random operator in (5.35) by its mean before squaring would delete this term and give a wrong quadratic coefficient.

5.9 Execute the second-displacement peel

Define the cross moment

$$S_{ab}^\ell := \mathbb{E}[s^\ell(a)h^\ell(b)], \quad S^0 = 0. \quad (5.43)$$

There are two terms in (5.27). The mixed matrix/feature term is, on the label branch,

$$\frac{1}{\sqrt{n}} \sum_j W_{ij}^\ell g_j^{\ell-1}(a) = \sum_c \lambda_c \delta_i^\ell(c) X_{n,ca}^{\ell-1}, \quad (5.44)$$

where

$$X_{n,ca}^{\ell-1} = \frac{1}{n} \sum_j h_j^{\ell-1}(c) g_j^{\ell-1}(a). \quad (5.45)$$

Its one-copy expectation is zero by readout parity, but that is not sufficient to discard it. Scalarizing two copies gives

$$\mathbb{E}[(X_{n,ca}^{\ell-1})^2] = \frac{1}{n^2} \sum_{j,k} \mathbb{E}[h_j(c) g_j(a) h_k(c) g_k(a)]. \quad (5.46)$$

The $j = k$ family has n assignments against the n^2 normalization and is $O(n^{-1})$. In the $j \neq k$ family, the disconnected readout contraction factorizes into two zero odd first moments. A connected contraction must either identify a current-layer row or attach one copy to the other by a Stein covariance carrying an extra n^{-1} . The lower-group boundary sums created by that attachment cannot restore the lost width: both copies must reconnect to the same current-layer backward field. Thus the connected family is also $O(n^{-1})$. Hence

$$X_{n,ca}^{\ell-1} = O_{L^2}(n^{-1/2}),$$

and (5.44) is negligible in the leading annealed coefficient after bounded-moment control.

It remains to peel

$$\frac{1}{\sqrt{n}} \sum_j W_{ij}^\ell s_j^{\ell-1}(a). \quad (5.47)$$

The target-row Stein response coefficient is

$$\mathbb{E} \left[\frac{\partial s^{\ell-1}(a)}{\partial \xi^{\ell-1}(c)} \right].$$

At this order $s^{\ell-1}$ is even under $v \mapsto -v$, whereas $\xi^{\ell-1}$ is odd. The differentiated random quantity is therefore odd, so its annealed response coefficient—not the random derivative itself—is zero. In the limiting representative-neuron law, the row-distinct Wick patterns leave a centered Gaussian base field Ω^ℓ with the covariance needed below:

$$\mathbb{E}[\Omega^\ell(a) Z_b^\ell] = S_{ab}^{\ell-1}. \quad (5.48)$$

Now use (5.26). Gaussian integration by parts in the jointly Gaussian pair (Ω^ℓ, Z^ℓ) gives

$$\begin{aligned} \mathbb{E}[\phi(Z_b^\ell) \phi'(Z_a^\ell) \Omega^\ell(a)] \\ = D_{ab}^\ell S_{ab}^{\ell-1} + \mathbb{E}[\phi(Z_b^\ell) \phi''(Z_a^\ell)] S_{aa}^{\ell-1}. \end{aligned} \quad (5.49)$$

The term $\frac{1}{2} \phi''(Z_a^\ell) u^\ell(a)^2$ has a fresh-field square and a response square. Combining them with (5.49) yields

$$\begin{aligned}
S_{ab}^\ell &= D_{ab}^\ell S_{ab}^{\ell-1} \\
&\quad + \mathbb{E}[\phi(Z_b^\ell) \phi''(Z_a^\ell)] \left(S_{aa}^{\ell-1} + \frac{1}{2} V_{aa}^{\ell-1} \right) \\
&\quad + \frac{1}{2} \sum_{c,d} \lambda_c \lambda_d T_{ac}^{\ell-1} T_{ad}^{\ell-1} P_{cd}^\ell \\
&\quad \times \mathbb{E}[\phi(Z_b^\ell) \phi''(Z_a^\ell) \phi'(Z_c^\ell) \phi'(Z_d^\ell)].
\end{aligned} \tag{5.50}$$

Every expectation in (5.42) and (5.50) is over the single B -dimensional vector

$$Z^\ell \sim N(0, G^{\ell-1}).$$

Finally, (5.28) becomes

$$C_{ab}^\ell = V_{ab}^\ell + S_{ab}^\ell + S_{ba}^\ell. \tag{5.51}$$

Thus the one-step hidden-Gram expansion is

$$\mathbb{E}[G_{n,ab}^{\ell,+}] = G_{ab}^\ell + \eta O(n^{-1}) + \eta^2 C_{ab}^\ell + o(\eta^2) + o_n(1). \tag{5.52}$$

5.10 Fully explicit first-layer coefficient

At layer 1, $V^0 = S^0 = 0$, $T^0 = G^0$, and $P_{cd}^1 = D_{cd}^2 D_{cd}^3$. Writing, only in the next display,

$$\phi_a = \phi(Z_a^1), \quad \phi'_a = \phi'(Z_a^1), \quad \phi''_a = \phi''(Z_a^1),$$

with $Z^1 \sim N(0, G^0)$, gives

$$\begin{aligned}
C_{ab}^1 &= \frac{4}{B^2} \sum_{c,d} y_c y_d P_{cd}^1 \left[G_{ac}^0 G_{bd}^0 \mathbb{E}[\phi'_a \phi'_b \phi'_c \phi'_d] \right. \\
&\quad + \frac{1}{2} G_{ac}^0 G_{ad}^0 \mathbb{E}[\phi_b \phi_a'' \phi'_c \phi'_d] \\
&\quad \left. + \frac{1}{2} G_{bc}^0 G_{bd}^0 \mathbb{E}[\phi_a \phi_b'' \phi'_c \phi'_d] \right].
\end{aligned} \tag{5.53}$$

This is already a completely explicit fixed-dimensional Gaussian integral. Layers 2 and 3 are obtained by evaluating (5.42), (5.50), and (5.51) in order.

5.11 Deep-linear audit of the one-step coefficient

Set $\phi(z) = z$. Then $D^\ell = P^\ell = 1$, $S^\ell = 0$, and every $G^\ell = G^0$. Define

$$\kappa = \frac{2}{B}, \quad \sigma_a = \sum_c G_{ac}^0 y_c.$$

The partial tangents are

$$T^0 = G^0, \quad T^1 = 2G^0, \quad T^2 = 3G^0.$$

The velocity recursion gives

$$C_{ab}^1 = \kappa^2 \sigma_a \sigma_b, \quad C_{ab}^2 = 5\kappa^2 \sigma_a \sigma_b, \quad C_{ab}^3 = 14\kappa^2 \sigma_a \sigma_b. \tag{5.54}$$

The nontrivial coefficient 5 can be checked directly. Put

$$M = \frac{1}{n} W^2 W^{2\top}.$$

The layer-2 velocity contains $I + M$, so

$$\frac{1}{n} \text{tr}(I + M)^2 \longrightarrow 1 + 2 \cdot 1 + 2 = 5,$$

because

$$\frac{1}{n} \text{tr} M \rightarrow 1, \quad \frac{1}{n} \text{tr} M^2 \rightarrow 2.$$

Replacing M by its entrywise mean I before forming the square would give 4, not 5. This solvable case directly verifies that the row-distinct fluctuation branch in (5.42) is essential.

6. Execution III: two gradient-descent steps

The second Euler step differs from simply doubling the first displacement because the gradient is recomputed at the parameters reached after step one. This section isolates that correction exactly, peels its readout, and gives the layer-by-layer state recursion needed to peel the hidden groups.

6.1 Exact two-step Taylor identity

Let the raw-coordinate μP vector field be

$$F(\theta) := -n\nabla \mathcal{J}(\theta). \quad (6.1)$$

Two Euler steps are

$$\theta^{(1)} = \theta + \eta F(\theta),$$

$$\theta^{(2)} = \theta^{(1)} + \eta F(\theta^{(1)}).$$

Taylor-expanding the second vector-field evaluation at finite width gives

$$\theta^{(2)} = \theta + 2\eta F + \eta^2 DF[F] + O(\eta^3). \quad (6.2)$$

For the observable

$$O(\theta) = G_{n,ab}^\ell(\theta),$$

define

$$L_{n,ab}^\ell = DO[F], \quad C_{n,ab}^\ell = \frac{1}{2}D^2O[F, F], \quad R_{n,ab}^\ell = DO[DF[F]]. \quad (6.3)$$

Then

$$G_{n,ab}^{\ell,(2)} = G_{n,ab}^\ell + 2\eta L_{n,ab}^\ell + \eta^2(4C_{n,ab}^\ell + R_{n,ab}^\ell) + O(\eta^3). \quad (6.4)$$

Subtracting the one-step expansion gives the change caused by the second update alone:

$$G_{n,ab}^{\ell,(2)} - G_{n,ab}^{\ell,(1)} = \eta L_{n,ab}^\ell + \eta^2(3C_{n,ab}^\ell + R_{n,ab}^\ell) + O(\eta^3). \quad (6.5)$$

The factors 4 and 3 are purely Taylor-combinatorial. The identity

$$D_F L_{n,ab}^\ell = 2C_{n,ab}^\ell + R_{n,ab}^\ell \quad (6.6)$$

is an independent check.

Section 5 proved $\mathbb{E}[L_{n,ab}^\ell] = O(n^{-1})$. Hence neither (6.4) nor (6.5) has a deterministic linear term after the sequential Taylor-then-width limit. Crucially, one may not set L_n^ℓ equal to its limiting value zero before differentiating it: (6.6) generally has a nonzero limit.

6.2 Isolate the new gradient correction

For squared loss,

$$\nabla^2 \mathcal{J} = \frac{2}{B} \sum_c \left(\nabla f_c \nabla f_c^\top + e_c \nabla^2 f_c \right). \quad (6.7)$$

Since $DF[F] = n^2 \nabla^2 \mathcal{J} \nabla \mathcal{J}$, substitution into (6.3) yields the exact finite-width identity

$$R_{n,ab}^\ell = \frac{4}{B^2} \sum_{c,d} e_d \left[\left(n \nabla G_{n,ab}^\ell \right)^\top \nabla f_c K_{n,cd} + e_c n^2 \left(\nabla G_{n,ab}^\ell \right)^\top \nabla^2 f_c \nabla f_d \right]. \quad (6.8)$$

The first line is the residual-change, or Gauss–Newton, branch. Its label part is odd in the centered readout and vanishes. Its surviving initialization-output branch is the same connected one-free-index-deficient family encountered in the linear Gram calculation, so it is $O(n^{-1})$.

In the second line, the label–label branch is even and survives, while mixed label/output branches are odd and the output–output branch is width-suppressed. Thus the leading correction is

$$R_{ab}^\ell = \frac{4}{B^2} \sum_{c,d} y_c y_d \lim_{n \rightarrow \infty} \mathbb{E} \left[n^2 \left(\nabla G_{n,ab}^\ell \right)^\top \nabla^2 f_c \nabla f_d \right]. \quad (6.9)$$

6.3 Execute the top readout peel exactly

Let θ and ψ denote individual raw parameter coordinates. Because a hidden Gram has no direct dependence on v , the contraction inside (6.9) is

$$n^2 \sum_{\theta \text{ hidden}} \sum_{\psi} \partial_{\theta} G_{n,ab}^\ell \partial_{\theta} \psi f_c \partial_{\psi} f_d. \quad (6.10)$$

There are two equality types for ψ .

If $\psi = v_s$, then

$$\partial_{\theta v_s} f_c = \frac{1}{n} \partial_{\theta} h_s^3(c), \quad \partial_{v_s} f_d = \frac{1}{n} h_s^3(d),$$

and the prefactor n^2 cancels the two readout normalizations.

If ψ is hidden, then

$$\partial_{\theta \psi} f_c = \frac{1}{n} \sum_r v_r \partial_{\theta \psi} h_r^3(c),$$

$$\partial_{\psi} f_d = \frac{1}{n} \sum_t v_t \partial_{\psi} h_t^3(d).$$

The exact readout Wick contraction $\mathbb{E}[v_r v_t] = \delta_{rt}$ again cancels the two factors $1/n$. Therefore

$$\begin{aligned} & \mathbb{E}_v \left[n^2 \left(\nabla G_{n,ab}^\ell \right)^\top \nabla^2 f_c \nabla f_d \middle| W^1, W^2, W^3 \right] \\ &= \sum_{\theta \text{ hidden}} \partial_{\theta} G_{n,ab}^\ell \left[\sum_r h_r^3(d) \partial_{\theta} h_r^3(c) \right. \\ & \quad \left. + \sum_{\psi \text{ hidden}} \sum_r \partial_{\theta \psi} h_r^3(c) \partial_{\psi} h_r^3(d) \right]. \end{aligned} \quad (6.11)$$

Equation (6.11) is the completely explicit boundary state handed from the readout group to group 3. No independence between a hidden matrix and the preactivations it creates has been used.

6.4 Exact differentiated forward and backward states

The Hessian contraction can be evaluated without writing a third-order parameter tensor. Differentiate the ordinary forward/backward program in the direction F .

Keep

$$u^\ell = D_F z^\ell, \quad g^\ell = D_F h^\ell = \phi'(z^\ell) u^\ell,$$

and define

$$\delta^\ell := D_F \delta^\ell. \quad (6.12)$$

At the top hidden layer the exact readout velocity and exact directional boundary are

$$\dot{v}_r = -\frac{2}{B} \sum_c e_c h_r^3(c),$$

$$\delta_r^3(a) = \dot{v}_r \phi'(z_r^3(a)) + v_r \phi''(z_r^3(a)) u_r^3(a). \quad (6.13)$$

Write a superscript (y) for the leading label-driven component of a directional state. Then

$$\dot{v}_r^{(y)} = \sum_c \lambda_c h_r^3(c),$$

$$\delta_r^{3,(y)}(a) = \left(\sum_c \lambda_c h_r^3(c) \right) \phi'(z_r^3(a)) + v_r \phi''(z_r^3(a)) u_r^{3,(y)}(a). \quad (6.13y)$$

For $\ell = 1, 2$, recall $\delta^\ell = \phi'(z^\ell) \xi^\ell$. Hence

$$\delta^\ell(a) = \phi''(z^\ell(a)) u^\ell(a) \xi^\ell(a) + \phi'(z^\ell(a)) \xi^\ell(a). \quad (6.14)$$

Its label-driven component satisfies

$$\delta^{\ell,(y)}(a) = \phi''(z^\ell(a)) u^{\ell,(y)}(a) \xi^\ell(a) + \phi'(z^\ell(a)) \xi^{\ell,(y)}(a). \quad (6.14y)$$

The incoming signal is a transpose multiplication, so its exact finite-width directional derivative is

$$\xi_j^\ell(a) = \frac{1}{\sqrt{n}} \sum_i \left[\dot{W}_{ij}^{\ell+1} \delta_i^{\ell+1}(a) + W_{ij}^{\ell+1} \delta_i^{\ell+1}(a) \right]. \quad (6.15)$$

For the label-driven component,

$$\dot{W}_{ij}^{\ell+1,(y)} = \frac{1}{\sqrt{n}} \sum_c \lambda_c \delta_i^{\ell+1}(c) h_j^\ell(c).$$

Peeling its contribution to the first term of (6.15) uses the already-computed backward Gram:

$$\begin{aligned} \frac{1}{\sqrt{n}} \sum_i \dot{W}_{ij}^{\ell+1,(y)} \delta_i^{\ell+1}(a) &= \sum_c \lambda_c h_j^\ell(c) \frac{1}{n} \sum_i \delta_i^{\ell+1}(c) \delta_i^{\ell+1}(a) \\ &\longrightarrow \sum_c \lambda_c h_j^\ell(c) \Pi_{ca}^{\ell+1}. \end{aligned} \quad (6.16)$$

The second term of (6.15) is a new backward transpose sum and must be peeled against every forward channel through the same matrix. Section 6.6 makes that operation explicit.

Now set

$$A := DF[F],$$

and define the correction-forward states

$$\bar{z}^\ell := D_A z^\ell, \quad \bar{h}^\ell := D_A h^\ell = \phi'(z^\ell) \bar{z}^\ell. \quad (6.17)$$

For concise denominators, let

$$\rho_1 = \sqrt{d_0}, \quad \rho_2 = \rho_3 = \sqrt{n}.$$

The exact differentiated vector field for a hidden matrix is

$$\begin{aligned} A_{W^\ell} = -\frac{2}{B\rho_\ell} \sum_c \Big[&\dot{f}_c \delta^\ell(c) h^{\ell-1}(c)^\top \\ &+ e_c \delta^\ell(c) h^{\ell-1}(c)^\top \\ &+ e_c \delta^\ell(c) g^{\ell-1}(c)^\top \Big]. \end{aligned} \quad (6.18)$$

Here $h^0(c) = x_c$ and $g^0 = 0$. In the first line of (6.18), the label-driven part of $\dot{f}_c \delta^\ell$ is odd in the readout and vanishes. Its initialization-output-driven part is even but belongs to the width-suppressed residual-change family identified below (6.8). Replacing e_c by $-y_c$ in the other two lines gives the leading even part

$$A_{W^\ell}^{\text{lead}} = \frac{1}{\rho_\ell} \sum_c \lambda_c \left[\dot{\delta}^{\ell,(y)}(c) h^{\ell-1}(c)^\top + \delta^\ell(c) g^{\ell-1,(y)}(c)^\top \right]. \quad (6.19)$$

The exact correction-forward recursion is

$$\bar{z}^1(a) = A_{W^1} \frac{x_a}{\sqrt{d_0}}, \quad (6.20)$$

and, for $\ell = 2, 3$,

$$\bar{z}^\ell(a) = \frac{1}{\sqrt{n}} W^\ell \bar{h}^{\ell-1}(a) + \frac{1}{\sqrt{n}} A_{W^\ell} h^{\ell-1}(a). \quad (6.21)$$

Finally, the correction to the hidden Gram is

$$R_{n,ab}^\ell = \frac{1}{n} \sum_i \left[\bar{h}_i^\ell(a) h_i^\ell(b) + h_i^\ell(a) \bar{h}_i^\ell(b) \right], \quad (6.22a)$$

When $\bar{h}^{\ell,\text{lead}}$ is built from A^{lead} , the formal leading mean-field replacement is

$$R_{ab}^\ell = \mathbb{E} \left[\bar{h}^{\ell,\text{lead}}(a) h^\ell(b) + h^\ell(a) \bar{h}^{\ell,\text{lead}}(b) \right]. \quad (6.22)$$

Equations (6.13), (6.14), (6.15), (6.18), and (6.20)–(6.22a) are exact when they are built from the full directional states. Equations (6.13y), (6.16), (6.19), and the later mean-field recursions (6.29)–(6.35) describe only the leading label–label branch. Equation (6.22) additionally requires joint row convergence and uniform integrability. Together these identities turn the Hessian contraction into ordinary products of explicit forward, backward, and differentiated states.

6.5 The multichannel Wick–Stein rule required at step two

At one step, a matrix was coupled mainly to one forward and one backward channel. Differentiation creates several channels through the same matrix, so the single-channel shorthand (3.1) must be generalized without dropping any covariance.

Let W be an iid standard Gaussian $n \times n$ matrix. Suppose the same matrix creates transpose channels

$$b_j^\alpha = \frac{1}{\sqrt{n}} \sum_i W_{ij} y_i^\alpha, \quad \alpha = 1, \dots, q. \quad (6.23)$$

If the source vectors y^α and the remaining lower-group variables are frozen with respect to the current matrix W , and Ψ_j is a smooth function of all $b_j^1, \dots, b_j^q$, then the exact one-weight identity gives

$$\mathbb{E}[W_{ij}\Psi_j] = \frac{1}{\sqrt{n}} \sum_{\alpha=1}^q y_i^\alpha \mathbb{E}[\partial_{b^\alpha} \Psi_j]. \quad (6.24)$$

After summing j and exposing equality patterns, the mean-field forward multiplication has the form

$$\frac{1}{\sqrt{n}} \sum_j W_{ij}\Psi_j \implies \Gamma_\Psi + \sum_{\alpha=1}^q y_i^\alpha \mathbb{E}[\partial_{b^\alpha} \Psi]. \quad (6.25)$$

The row-distinct patterns form the centered Gaussian field Γ_Ψ . It is generated jointly with every other forward sum through W ; for two integrands Ψ and $\tilde{\Psi}$,

$$\mathbb{E}[\Gamma_\Psi(a)\Gamma_{\tilde{\Psi}}(b)] = \mathbb{E}[\Psi(a)\tilde{\Psi}(b)]. \quad (6.26)$$

All cross-covariances with previously created fields using the same matrix must also be retained. The transpose version is symmetric. If

$$z_i^\alpha = \frac{1}{\sqrt{n}} \sum_j W_{ij}x_j^\alpha,$$

and Υ_i depends on every z_i^α , then

$$\frac{1}{\sqrt{n}} \sum_i W_{ij}\Upsilon_i \implies \Xi_\Upsilon + \sum_{\alpha=1}^q x_j^\alpha \mathbb{E}[\partial_{z^\alpha} \Upsilon], \quad (6.27)$$

with joint fresh-field covariance

$$\mathbb{E}[\Xi_\Upsilon(a)\Xi_\Omega(b)] = \mathbb{E}[\Upsilon(a)\Omega(b)]. \quad (6.28)$$

Equations (6.25) and (6.27) are not independence substitutions. Their response terms are precisely the accumulated Stein/Onsager corrections; their fresh Gaussian terms are the surviving off-diagonal Wick families.

Equations (6.24)–(6.27) apply directly only when their channel sources are frozen with respect to the current matrix. If a source such as y^α or x^α itself depends on W , that dependence must first be exposed as additional forward or transpose channels, and the full integration-by-parts derivative must include it.

For a rigorous use of these rules, the list of channels must therefore be complete. A derivative such as $\partial_{z^\alpha} \Upsilon$ is a total derivative through all deterministic response pieces already built from that channel. If a fresh field made earlier with the same matrix reappears, it must be registered as another jointly generated Gaussian base channel rather than declared independent.

6.6 Peel the differentiated backward pass

Apply (6.27) to the label-driven component of the second term of (6.15). The visible forward channels through $W^{\ell+1}$ are the ordinary preactivation $z^{\ell+1}$, sourced by h^ℓ , and its first label tangent $u^{\ell+1,(y)}$, sourced by $g^{\ell,(y)}$. Thus the first two response terms are

$$\sum_c h_j^\ell(c) \mathbb{E} \left[\frac{\partial \delta^{\ell+1,(y)}(a)}{\partial Z_c^{\ell+1}} \right]$$

and

$$\sum_c g_j^{\ell,(y)}(c) \mathbb{E} \left[\frac{\partial \delta^{\ell+1,(y)}(a)}{\partial u^{\ell+1,(y)}(c)} \right].$$

Together with the direct term (6.16), the peeled form is

$$\begin{aligned} \dot{\xi}^{\ell,(y)}(a) = & \sum_c \lambda_c h^\ell(c) \Pi_{ca}^{\ell+1} + \Xi^{\ell,(y)}(a) \\ & + \sum_c h^\ell(c) \mathbb{E} \left[\partial_{Z_c^{\ell+1}} \delta^{\ell+1,(y)}(a) \right] \\ & + \sum_c g^{\ell,(y)}(c) \mathbb{E} \left[\partial_{u^{\ell+1,(y)}(c)} \delta^{\ell+1,(y)}(a) \right] + \text{responses of any additional registered channel.} \end{aligned} \quad (6.29)$$

The Gaussian base field $\Xi^{\ell,(y)}$ is centered. It is generated jointly with the ordinary incoming field ξ^ℓ and all other transpose sums through the same matrix. In particular,

$$\mathbb{E}[\Xi^{\ell,(y)}(a)\Xi^{\ell,(y)}(b)] = \mathbb{E}[\delta^{\ell+1,(y)}(a)\delta^{\ell+1,(y)}(b)],$$

$$\mathbb{E}[\Xi^{\ell,(y)}(a)\xi^\ell(b)] = \mathbb{E}[\delta^{\ell+1,(y)}(a)\delta^{\ell+1}(b)]. \quad (6.30)$$

Equations (6.13y), (6.14y), and (6.29), evaluated in the order $3 \rightarrow 2 \rightarrow 1$, define the formal multichannel peel of the label-driven differentiated

backward pass once a complete channel registry and covariance block are supplied. The placeholder in (6.29) makes this a schema, not a fully enumerated nonlinear closed recursion.

6.7 Peel the correction-forward pass

In the leading branch of (6.21), the term $W^\ell \bar{h}^{\ell-1, \text{lead}} / \sqrt{n}$ is a forward multiplication. The current matrix also generates the transpose channels $\xi^{\ell-1}$ and $\xi^{\ell-1, (y)}$, sourced respectively by δ^ℓ and $\delta^{\ell, (y)}$. Applying (6.25) gives

$$\begin{aligned} \frac{1}{\sqrt{n}} W^\ell \bar{h}^{\ell-1, \text{lead}}(a) &\Longrightarrow \bar{\Gamma}^{\ell, \text{lead}}(a) \\ &+ \sum_c \delta^\ell(c) \mathbb{E}[\partial_{\xi^{\ell-1}(c)} \bar{h}^{\ell-1, \text{lead}}(a)] \\ &+ \sum_c \delta^{\ell, (y)}(c) \mathbb{E}[\partial_{\xi^{\ell-1, (y)}(c)} \bar{h}^{\ell-1, \text{lead}}(a)] \\ &+ \text{responses of any additional registered channel.} \end{aligned} \quad (6.31)$$

The fresh Gaussian base field $\bar{\Gamma}^{\ell, \text{lead}}$ must be generated jointly with $Z^\ell, u^{\ell, (y)}$, and every other forward sum through W^ℓ ; its covariance and all cross-covariances follow from (6.26).

The direct $A_{W^\ell} h^{\ell-1}$ term in (6.21) gives

$$\sum_c \lambda_c \delta^{\ell, (y)}(c) G_{ac}^{\ell-1} + \sum_c \lambda_c \delta^\ell(c) \frac{1}{n} \sum_j g_j^{\ell-1, (y)}(c) h_j^{\ell-1}(a). \quad (6.32)$$

The second empirical average is odd in the readout. Its disconnected limit is zero, and a connected two-copy peel is width-suppressed. Therefore the leading correction-forward recursion is

$$\bar{z}^{\ell, \text{lead}}(a) = \text{ForwardPeel}_\ell[\bar{h}^{\ell-1, \text{lead}}(a)] + \sum_c \lambda_c G_{ac}^{\ell-1} \delta^{\ell, (y)}(c), \quad (6.33)$$

where ForwardPeel means the right-hand side of (6.31) after a complete channel registry has been supplied, including every response and joint covariance. Since that registry is not fully enumerated here, (6.33) is a formal schema. For layer 1 there is no propagated term, so

$$\bar{z}^{1, \text{lead}}(a) = \sum_c \lambda_c G_{ac}^0 \delta^{1, (y)}(c). \quad (6.34)$$

With $\bar{h}^{\ell, \text{lead}} = \phi'(z^\ell) \bar{z}^{\ell, \text{lead}}$, substitution into (6.22) represents R^ℓ as fixed-dimensional Gaussian expectations. For example, once the differentiated-backward channel law has been supplied, the layer-1 endpoint is

$$\begin{aligned} R_{ab}^1 = \sum_c \lambda_c \Big[&G_{ac}^0 \mathbb{E}[\phi(Z_b^1) \phi'(Z_a^1) \delta^{1, (y)}(c)] \\ &+ G_{bc}^0 \mathbb{E}[\phi(Z_a^1) \phi'(Z_b^1) \delta^{1, (y)}(c)] \Big]. \end{aligned} \quad (6.35)$$

The nonlinear variables inside (6.35) are explicit functions of the Gaussian base channels produced by the preceding forward, backward, first-tangent, and differentiated-backward peels. At fixed derivative order the Gaussian base-channel count is finite, so each layer requires an expectation over a fixed multiple qB of batch coordinates, independent of width. It need not be only one B -dimensional vector.

6.8 Execution schema for the general nonlinear correction

For clarity, the reusable two-step computation is the following finite-state program schema.

1. **Forward pass:** compute G^0, G^1, G^2, G^3 and retain representative Gaussian preactivations Z^1, Z^2, Z^3 .
2. **Ordinary backward peel:** compute ξ^3, δ^3 , then ξ^2, δ^2 , then ξ^1, δ^1 , retaining their complete joint covariances. This gives P^ℓ and Π^ℓ .
3. **First-tangent peel:** compute $u^{1, (y)}, g^{1, (y)}$, then $u^{2, (y)}, g^{2, (y)}$, then $u^{3, (y)}, g^{3, (y)}$ using the diagonal Stein responses and fresh off-diagonal Gaussian base fields in (5.39).
4. **Differentiated-backward peel:** start with the label-leading top boundary (6.13y). Then use (6.14y), (6.16), and the multichannel schema (6.29) for groups $3 \rightarrow 2 \rightarrow 1$.
5. **Correction-forward peel:** compute $\bar{z}^{1, \text{lead}}, \bar{h}^{1, \text{lead}}$ from (6.34), followed by layers 2 and 3 using (6.31)–(6.33). At each matrix, generate all fresh Gaussian base fields jointly with fields previously made from that matrix.
6. **Observable readout:** evaluate (6.22). Each entry is the expectation of an explicit function of the finite Gaussian base channels accumulated above.

The finite-width directional equations and the underlying multichannel one-weight integration-by-parts identity (6.24), under its frozen-source condition, are exact. The Gaussian-limit arrows (6.25) and (6.27) are formal mean-field conclusions under the assumptions in Section 2.7. The claim is that the Gaussian base channels converge jointly and every retained forward, backward, and differentiated state is an explicit nonlinear function of them—not that the entire state is jointly Gaussian. A theorem must additionally prove the leave-one-out estimates, uniform moment bounds, source-channel completeness, and joint convergence for the registered Gaussian base channels. The finite count q may depend on derivative order and on the chosen depth/state construction, but not on width. This qualification matters more at two steps than at one because differentiated fresh fields can be correlated with several earlier channels through the same matrix.

6.9 Two-step result

Subject to completion and convergence of the nonlinear channel registry in Section 6.8, the formal mean-field closure is, coefficient by coefficient,

$$\mathbb{E}[G_{n,ab}^{\ell,(2)}] = G_{ab}^{\ell} + \eta O(n^{-1}) + \eta^2(4C_{ab}^{\ell} + R_{ab}^{\ell}) + o(\eta^2) + o_n(1). \quad (6.36)$$

For the second update alone,

$$\mathbb{E}[G_{n,ab}^{\ell,(2)} - G_{n,ab}^{\ell,(1)}] = \eta O(n^{-1}) + \eta^2(3C_{ab}^{\ell} + R_{ab}^{\ell}) + o(\eta^2) + o_n(1). \quad (6.37)$$

The expectations of both linear coefficients are $O(n^{-1})$ before the width limit; the random coefficients themselves need not be. The term C^{ℓ} is the one-step quadratic coefficient from Section 5, and R^{ℓ} is due entirely to recomputing the gradient. Equations (6.36)–(6.37) are formal for a general nonlinear activation and verified explicitly in the deep-linear specialization below.

6.10 Deep-linear audit of the two-step correction

Again set $\phi(z) = z$, let

$$\kappa = \frac{2}{B}, \quad \sigma = G^0 y,$$

and write

$$\tau(A) = \frac{1}{n} \operatorname{tr} A.$$

The one-step coefficients are

$$C_{ab}^{\ell} = (1, 5, 14)_{\ell} \kappa^2 \sigma_a \sigma_b.$$

For the correction term define

$$M_2 = \frac{1}{n} W^{2\top} W^2, \quad M_3 = \frac{1}{n} W^{3\top} W^3, \quad M_{23} = \frac{1}{n} W^{2\top} M_3 W^2. \quad (6.38)$$

Gaussian Wick contraction gives

$$\begin{aligned} \tau(M_2) &= \tau(M_3) = \tau(M_{23}) = 1, \\ \tau(M_2^2) &= \tau(M_3^2) = 2, \\ \tau(M_{23}M_2) &= 2, \quad \tau(M_{23}^2) = 3. \end{aligned} \quad (6.39)$$

For example, expanding

$$\tau(M_2^2) = \frac{1}{n^3} \sum_{a,b,i,j} W_{ia}^2 W_{ib}^2 W_{jb}^2 W_{ja}^2$$

produces three Wick pairings. Two retain three free indices and contribute 1 each; the crossing pairing retains only two and is $O(n^{-1})$. Hence $\tau(M_2^2) \rightarrow 2$.

Because M_{23} contains the independent Wishart matrix M_3 only once in $\tau(M_{23}M_2)$, conditioning on W^2 and using $\mathbb{E}[M_3] = I$ reduces it to $\tau(M_2^2)$, giving 2.

For $\tau(M_{23}^2)$, put $\tilde{M}_2 = W^2 W^{2\top} / n$. Cyclicity gives

$$\tau(M_{23}^2) = \tau(M_3 \tilde{M}_2 M_3 \tilde{M}_2).$$

For fixed symmetric $\tilde{M}_2$, the exact second moment of the Wishart entries is

$$\mathbb{E}[(M_3)_{ij}(M_3)_{kl}] = \delta_{ij}\delta_{kl} + \frac{1}{n}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

Substitution into the normalized trace yields, at leading order,

$$\mathbb{E}_{M_3}[\tau(M_3 \tilde{M}_2 M_3 \tilde{M}_2)] = \tau(\tilde{M}_2^2) + \tau(\tilde{M}_2)^2 + o(1) \rightarrow 2 + 1 = 3.$$

To connect these traces to the differentiated-state program, define the label-aggregated forward vectors

$$a^{\ell} := \sum_c \lambda_c h^{\ell}(c),$$

and the ordinary linear backward vectors

$$b^3 = v, \quad b^2 = \frac{1}{\sqrt{n}} W^{3\top} b^3, \quad b^1 = \frac{1}{\sqrt{n}} W^{2\top} b^2.$$

Because

$$a^2 = \frac{1}{\sqrt{n}} W^2 a^1, \quad a^3 = \frac{1}{\sqrt{n}} W^3 a^2,$$

differentiating the backward pass along the label-driven first update gives

$$\dot{b}^3 = a^3,$$

$$\dot{b}^2 = \frac{\|b^3\|^2}{n} a^2 + \frac{1}{\sqrt{n}} W^{3\top} a^3 \implies (I + M_3) a^2,$$

and

$$\dot{b}^1 = \frac{\|b^2\|^2}{n} a^1 + \frac{1}{\sqrt{n}} W^{2\top} \dot{b}^2 \implies (I + M_2 + M_{23}) a^1. \quad (6.39a)$$

The first term in each line comes from differentiating the corresponding weight matrix; the remaining terms propagate the differentiated backward vector. Inserting (6.39a) into the correction-forward recursion (6.19)–(6.21) gives the following one-sided Gram contractions.

The one-sided correction contractions at layers 1, 2, and 3 are respectively

$$3 = \tau(I + M_2 + M_{23}),$$

$$7 = \tau[M_2(I + M_2 + M_{23})] + \tau(I + M_3) = (1 + 2 + 2) + (1 + 1),$$

and

$$\begin{aligned} 10 &= \tau[M_{23}(I + M_2 + M_{23})] + \tau[M_3(I + M_3)] + 1 \\ &= (1 + 2 + 3) + (1 + 2) + 1. \end{aligned}$$

The Gram has two endpoints, so symmetrization doubles these values:

$$\boxed{R_{ab}^1 = 6\kappa^2 \sigma_a \sigma_b, \quad R_{ab}^2 = 14\kappa^2 \sigma_a \sigma_b, \quad R_{ab}^3 = 20\kappa^2 \sigma_a \sigma_b.} \quad (6.40)$$

Therefore the cumulative two-step coefficients are

$$\boxed{(4C^\ell + R^\ell)_{ab} = (10, 34, 76)_\ell \kappa^2 \sigma_a \sigma_b,} \quad (6.41)$$

and the second-step increment coefficients are

$$\boxed{(3C^\ell + R^\ell)_{ab} = (9, 29, 62)_\ell \kappa^2 \sigma_a \sigma_b.} \quad (6.42)$$

This audit checks the factors 4 and 3, the gradient-correction sign, and the off-diagonal width patterns. It also distinguishes two Euler steps from gradient flow evaluated at time 2η : the latter would have quadratic coefficient $4C^\ell + 2R^\ell$.

7. What this case study changes in the peeling program

The three executions support the core peeling architecture, but they sharpen several statements that should be part of any theorem-level formulation.

1. **The raw/effective coordinate convention must be fixed before peeling.** In the raw convention used here,

$$\text{Cov}(W_{rp}^3, z_r^3(c) \mid h^2) = \frac{h_p^2(c)}{\sqrt{n}}.$$

Writing $h_p^2(c)/n$ would correspond to a different effective-weight convention and would corrupt width counting.

2. **Exact conditional Gaussian integration precedes deterministic Gram replacement.** A singled-out row may also occur in the empirical covariance of its layer. The safe proof uses leave-one-out or leave-two-out covariances, performs Wick–Stein first, and only then sends the empirical covariance to its deterministic limit.
3. **Stein-created lower-group factors remain boundary data.** For example, $h_p^2(c)h_q^2(d)$ created while peeling group 3 belongs to group 2. It cannot be averaged or summed out during the group-3 peel.
4. **An off-diagonal family cannot be removed from its one-copy mean.** In the backward kernel it is subleading after complete counting; in the quadratic feature coefficient it survives as the fresh field Γ^ℓ . The observable and the number of copies determine the answer.
5. **Expectation and concentration are separate proof obligations.** One-copy peeling identifies an annealed limit. Replacing empirical kernels inside later products requires a two-copy or higher-copy concentration argument plus uniform integrability.
6. **A limiting zero cannot be differentiated as zero.** Although $L_n^\ell \rightarrow 0$, its directional derivative contains the nonzero terms $2C^\ell + R^\ell$.
7. **Higher training steps require derivative-state augmentation.** At step two, the state must include $u, g, \delta, \xi, \bar{z}, \bar{h}$ in addition to the ordinary forward and backward variables. At higher Taylor order this becomes a finite jet of differentiated states.
8. **The Gaussian integration dimension is a fixed multiple of batch size.** A primitive forward kernel uses one B -dimensional Gaussian vector. A higher-order calculation uses a finite collection of jointly Gaussian base vectors, and the retained nonlinear states are explicit functions of

them. Their total Gaussian dimension is qB . The number q may depend on derivative order and on the chosen depth/state construction, but not on width.

9. **An $O(L)$ recursion still requires depth-uniform finite-state closure.** The intended algorithm traverses depth a constant number of times at fixed jet order, but the nonlinear registry is still schematic and its base-channel count may depend on depth. Linear complexity in L is therefore a target conclusion, not yet an established result. It requires proving that the registered state size is bounded independently of depth at fixed derivative order.
10. **The nonlinear two-step closure still carries theorem obligations.** The exact finite-width directional identities are complete, but the displayed nonlinear mean-field channel registry is schematic. A rigorous theorem must enumerate the registry and prove joint convergence of its Gaussian base channels while tracking every cross-covariance; the deep-linear audit alone does not prove the general nonlinear closure.

8. Final consolidated statement

For this three-hidden-layer μP network, the mean-field peeling calculation gives the following coefficientwise results, subject to the concentration and joint-state obligations stated above:

$$\Pi^\ell = D^\ell \odot D^{\ell+1} \odot \dots \odot D^3,$$

$$\lim_{n \rightarrow \infty} [\eta] \mathbb{E}[\ell_a^+] = -\frac{4}{B} y_a \sum_c K_{ac} y_c,$$

$$\lim_{n \rightarrow \infty} [\eta] \mathbb{E}[G_{n,ab}^{\ell,+}] = 0,$$

$$\lim_{n \rightarrow \infty} [\eta^2] \mathbb{E}[G_{n,ab}^{\ell,+}] = C_{ab}^\ell,$$

and

$$\lim_{n \rightarrow \infty} [\eta^2] \mathbb{E}[G_{n,ab}^{\ell,(2)}] = 4C_{ab}^\ell + R_{ab}^\ell.$$

The backward kernels and one-step coefficient C^ℓ close through explicit fixed-batch Gaussian expectations under the stated joint-CLT assumptions. The two-step correction R^ℓ is represented by the exact differentiated-state identities and formally reduced through the multichannel Gaussian-peeling schema in Section 6. Its general nonlinear registry and convergence remain proof obligations, while its full deep-linear specialization is explicitly verified in (6.40)–(6.42).

Appendix A. General fixed-depth backward-kernel template

The main text uses the raw-coordinate specialization with three hidden layers, unit hidden variances, and a centered $1/n$ readout. This appendix preserves the more general normalization that motivated that execution. It is a fixed-depth template supported by the explicit three-layer audit; a complete all-depth diagram induction and concentration theorem remain proof obligations.

Let hidden layers be $1, \dots, H$, all of width n , and set $L := H + 1$ for the scalar readout group. Use effective hidden weights

$$A_{ij}^\ell \sim N\left(0, \frac{s_\ell^2}{n}\right),$$

and the scalar readout

$$f_a = s_L n^{-\alpha} \sum_{i=1}^n a_i h_i^H(a), \quad a_i \stackrel{\text{iid}}{\sim} N(0, 1). \quad (\text{A.1})$$

The choices $\alpha = 1/2$ and $\alpha = 1$ give, respectively, a standard random-output/NTK-size readout and the centered mean-field/ μP -size readout used in the main calculation.

Define

$$g_i^{\ell,a} := \frac{\partial f_a}{\partial z_i^\ell(a)}, \quad \beta_i^{\ell,a} := n^\alpha g_i^{\ell,a}.$$

Then

$$\beta_i^{H,a} = s_L a_i \phi'(z_i^{H,a}), \quad (\text{A.2})$$

and, for $\ell < H$,

$$\beta_i^{\ell,a} = \phi'(z_i^{\ell,a}) \sum_{r=1}^n A_{ri}^{\ell+1} \beta_r^{\ell+1,a}. \quad (\text{A.3})$$

The normalized backward kernel is

$$\widehat{\Pi}_{n,ab}^\ell := \frac{1}{n} \sum_i \beta_i^{\ell,a} \beta_i^{\ell,b} = n^{2\alpha-1} \sum_i g_i^{\ell,a} g_i^{\ell,b}. \quad (\text{A.4})$$

Thus (A.4) is the ordinary backward inner product for $\alpha = 1/2$; for $\alpha = 1$, the ordinary inner product is $n^{-1} \widehat{\Pi}_{n,ab}^\ell$.

Let $Z^\ell \sim N(0, Q^\ell)$ and set

$$D_{ab}^\ell := \mathbb{E}[\phi'(Z_a^\ell)\phi'(Z_b^\ell)]. \quad (\text{A.5})$$

For a hidden transition, the corresponding derivative-transport coefficient is

$$\dot{Q}_{ab}^\ell := s_{\ell+1}^2 D_{ab}^\ell. \quad (\text{A.5a})$$

The readout contraction gives the boundary

$$\widehat{\Pi}_{n,ab}^H = \frac{s_L^2}{n} \sum_i a_i^2 \phi'(z_i^{H,a}) \phi'(z_i^{H,b}) \longrightarrow s_L^2 D_{ab}^H. \quad (\text{A.6})$$

For a lower transition, put

$$u_i^{\ell,ab} = \phi'(z_i^{\ell,a}) \phi'(z_i^{\ell,b}).$$

Recursive scalarization gives the exact expression

$$\widehat{\Pi}_{n,ab}^\ell = \frac{1}{n} \sum_{i,r,s} u_i^{\ell,ab} A_{ri}^{\ell+1} A_{si}^{\ell+1} \beta_r^{\ell+1,a} \beta_s^{\ell+1,b}. \quad (\text{A.7})$$

If $v_{\ell+1} = s_{\ell+1}^2/n$, Gaussian integration by parts gives, for a smooth F of the current matrix,

$$\mathbb{E}[A_{ri}^{\ell+1} A_{si}^{\ell+1} F] = v_{\ell+1} \delta_{rs} \mathbb{E}[F] + v_{\ell+1}^2 \mathbb{E}[\partial_{A_{ri}^{\ell+1}} \partial_{A_{si}^{\ell+1}} F]. \quad (\text{A.8})$$

Because the effective coordinate satisfies

$$\partial_{A_{ri}^{\ell+1}} = \sum_{c=1}^B h_i^{\ell,c} \partial_{z_r^{\ell+1,c}},$$

(A.8), applied to $F = \beta_r^{\ell+1,a} \beta_s^{\ell+1,b}$, yields

$$\mathbb{E}_{A^{\ell+1}}[\widehat{\Pi}_{n,ab}^\ell] = s_{\ell+1}^2 \widehat{D}_{n,ab}^\ell \mathbb{E}_{A^{\ell+1}}[\widehat{\Pi}_{n,ab}^{\ell+1}] + R_{n,ab}^\ell \quad (\text{A.9})$$

where

$$\widehat{D}_{n,ab}^\ell = \frac{1}{n} \sum_i u_i^{\ell,ab}$$

and

$$\begin{aligned} R_{n,ab}^\ell &= \frac{s_{\ell+1}^4}{n^3} \sum_{i,r,s} \sum_{c,d=1}^B u_i^{\ell,ab} h_i^{\ell,c} h_i^{\ell,d} \\ &\quad \times \mathbb{E} \left[\partial_{z_r^{\ell+1,c}} \partial_{z_s^{\ell+1,d}} (\beta_r^{\ell+1,a} \beta_s^{\ell+1,b}) \right]. \end{aligned} \quad (\text{A.10})$$

Equations (A.9)–(A.10) are exact after the stated conditional expectation. They display the Wick branch and the complete Onsager remainder without an independence substitution. In the top transition the explicit same-coordinate identity is

$$\begin{aligned} &\mathbb{E}[(A_{ri}^H)^2 \phi'(Z_r^H(a)) \phi'(Z_r^H(b)) \mid \mathcal{F}_{H-1}] \\ &= \frac{s_H^2}{n} \mathbb{E}[\phi'(Z_a) \phi'(Z_b)] \\ &\quad + \frac{s_H^4}{n^2} \sum_{c,d} h_i^{H-1,c} h_i^{H-1,d} \mathbb{E}[\partial_{cd} \{\phi'(Z_a) \phi'(Z_b)\}], \end{aligned} \quad (\text{A.11})$$

where $Z \sim N(0, Q_n^H)$. The second line after the leading term contains the $\phi''' \phi'$, $\phi'' \phi''$, and $\phi' \phi'''$ integrands and is $O(n^{-1})$ after the normalized row sum under the stated bounded-moment assumptions.

Substitution of (A.11) gives the explicit top-transition decomposition

$$\begin{aligned} \mathbb{E}[\widehat{\Pi}_{n,ab}^{H-1} \mid \mathcal{F}_{H-1}] &= s_H^2 \widehat{D}_{n,ab}^{H-1} \left[s_L^2 \mathbb{E}_{Z \sim N(0, Q_n^H)} [\phi'(Z_a) \phi'(Z_b)] \right] \\ &\quad + \frac{s_L^2 s_H^4}{n^2} \sum_i u_i^{H-1,ab} \sum_{c,d} h_i^{H-1,c} h_i^{H-1,d} \\ &\quad \times \mathbb{E}[\partial_{cd} \{\phi'(Z_a) \phi'(Z_b)\}]. \end{aligned} \quad (\text{A.11a})$$

The second term is an explicit, nonzero Onsager correction before the width limit. It is subleading for this normalized backward expectation under the stated assumptions.

Unrolling (A.3) gives the two-path representation used for the general ladder picture:

$$\beta_{i_\ell}^{\ell,a} = s_L \sum_{i_{\ell+1}, \dots, i_H} a_{i_H} \left(\prod_{t=\ell}^H \phi'(z_{i_t}^{\ell,a}) \right) \left(\prod_{t=\ell+1}^H A_{i_t i_{t-1}}^t \right). \quad (\text{A.11b})$$

In the leading ladder, the readout contraction identifies the terminal path indices and each hidden weight pairing identifies the two path indices at the next level. Every shared hidden index supplies one factor n , while every paired effective hidden weight supplies one factor n^{-1} ; these powers cancel layer by layer and leave the derivative moments (A.5). A Stein attachment can release a Wick equality but carries an additional inverse width and changes the lower boundary signature, so its final order must be decided only after the paths reconnect or all lower sums are exposed.

For a centered readout, a nominally leading disconnected split also vanishes because each isolated backward branch has zero first moment. This suppression is specific to this normalized backward second-moment observable; it is not a universal pseudo-independence statement for backward or higher-derivative objects.

The formal arbitrary-fixed-depth ladder recursion is

$$\Pi^H = s_L^2 D^H, \quad \Pi^\ell = s_{\ell+1}^2 D^\ell \odot \Pi^{\ell+1}, \quad 1 \leq \ell < H, \quad (\text{A.12})$$

or entrywise

$$\Pi_{ab}^\ell = \left(\prod_{r=\ell+1}^L s_r^2 \right) \left(\prod_{t=\ell}^H D_{ab}^t \right). \quad (\text{A.13})$$

The main text explicitly verifies the required suppression of all Stein branches for $H = 3$ at expectation level. Equations (A.12)–(A.13) are the general ladder template; a complete induction over every arbitrary-depth boundary pattern has not been supplied here.

Likewise, the expected concentration mechanism is a four-path expansion. The disconnected pair of ladders reproduces

$$(\mathbb{E} \widehat{\Pi}_{n,ab}^\ell)^2,$$

whereas each connected cross-copy pattern is expected to lose at least one free neuron block. Establishing

$$\text{Var}(\widehat{\Pi}_{n,ab}^\ell) = O(n^{-1})$$

requires that complete enumeration and is not supplied by the two-path calculation. Under such a theorem, the deterministic Stein bias is typically $O(n^{-1})$, while finite-width random fluctuations are typically $O_p(n^{-1/2})$.

For the sample loss

$$\mathcal{L}_a = (f_a - y_a)^2,$$

the normalized loss adjoint obeys the exact identity

$$n^\alpha \frac{\partial \mathcal{L}_a}{\partial z_i^\ell(a)} = 2(f_a - y_a) \beta_{i_\ell}^{\ell,a},$$

and therefore

$$\widehat{\Pi}_{n,ab}^{\ell,\text{loss}} = 4(f_a - y_a)(f_b - y_b) \widehat{\Pi}_{n,ab}^\ell. \quad (\text{A.14})$$

For the centered $\alpha = 1$ readout, $f_a \rightarrow 0$ and the limiting loss-adjoint kernel is $4y_a y_b \Pi_{ab}^\ell$. For $\alpha = 1/2$, the output has a nondegenerate Gaussian-process limit F , so the loss-adjoint kernel generally retains the random multiplier

$$4(F_a - y_a)(F_b - y_b) \Pi_{ab}^\ell.$$

If F is centered with covariance K^f , its annealed expectation is

$$4(K_{ab}^f + y_a y_b) \Pi_{ab}^\ell. \quad (\text{A.15})$$

The backward kernel here is a second-moment object. It is not the mixed first moment $M_{k,\ell}$ from the historical NTK sample. With a declared parameter coordinate and optimizer metric, a layerwise tangent-kernel contribution is a forward activation Gram multiplied entrywise by the corresponding backward kernel, together with the appropriate scaling factor.

Appendix B. Complete two-sample five-branch backward audit

Section 4 gives the full fixed-batch derivation and branch-order count. This appendix records the explicit two-sample branch formula that is useful for checking a symbolic implementation. It adds no new independence assumption.

Fix two samples a, b . For $\ell = 1, 2, 3$, let $\mathbb{E}_\ell$ denote expectation under $Z^\ell \sim N(0, G^{\ell-1})$ and define

$$D_\ell = \mathbb{E}_\ell[\phi'(Z_a) \phi'(Z_b)],$$

$$C_\ell(c, d) = \mathbb{E}_\ell[\partial_c \partial_d \{\phi'(Z_a) \phi'(Z_b)\}]. \quad (\text{B.1})$$

The only nonzero second derivatives in (B.1) are generated by occurrences of a and b . When $a \neq b$, they are explicitly

$$\begin{array}{c|c}
(c, d) & \partial_c \partial_d \{\phi'(Z_a) \phi'(Z_b)\} \\
\hline
(a, a) & \phi'''(Z_a) \phi'(Z_b) \\
(a, b) & \phi''(Z_a) \phi''(Z_b) \\
(b, a) & \phi''(Z_a) \phi''(Z_b) \\
(b, b) & \phi'(Z_a) \phi'''(Z_b).
\end{array} \quad (\text{B.2})$$

If $a = b$, coincident factor occurrences must be summed. For example,

$$\partial_{aa}[\phi'(Z_a)^2] = 2\phi'''(Z_a)\phi'(Z_a) + 2\phi''(Z_a)^2. \quad (\text{B.2a})$$

The occurrence formula (B.5) below handles both distinct and coincident sample labels without a separate case convention.

For auxiliary sample labels c, d , set

$$P_{cd}(Z) = \phi'(Z_a)\phi'(Z_b)\phi(Z_c)\phi(Z_d),$$

$$L_c(Z) = \phi'(Z_a)\phi(Z_c), \quad R_d(Z) = \phi'(Z_b)\phi(Z_d). \quad (\text{B.3})$$

The one-row first derivatives are fully explicit. For example,

$$\partial_u L_c(Z) = \mathbf{1}_{u=a} \phi''(Z_a) \phi(Z_c) + \mathbf{1}_{u=c} \phi'(Z_a) \phi'(Z_c), \quad (\text{B.4})$$

with the analogous formula for R_d after replacing a by b .

To expand the second derivative of P_{cd} without hiding product-rule terms, regard its four factor occurrences as

$$(a, 1), \quad (b, 1), \quad (c, 0), \quad (d, 0),$$

where the second coordinate is the derivative order and $\phi^{(0)} = \phi$. If these pairs are denoted locally by (r_j, m_j) , then

$$\begin{aligned}
\partial_u \partial_w P_{cd}(Z) &= \sum_{j=1}^4 \mathbf{1}_{r_j=u} \mathbf{1}_{r_j=w} \phi^{(m_j+2)}(Z_{r_j}) \prod_{k \neq j} \phi^{(m_k)}(Z_{r_k}) \\
&\quad + \sum_{j \neq k} \mathbf{1}_{r_j=u} \mathbf{1}_{r_k=w} \phi^{(m_j+1)}(Z_{r_j}) \phi^{(m_k+1)}(Z_{r_k}) \prod_{l \neq j, k} \phi^{(m_l)}(Z_{r_l}).
\end{aligned} \quad (\text{B.5})$$

The occurrence notation handles repeated labels such as $c = a$ correctly.

After the readout peel, exact group-3 and group-2 Wick–Stein eliminations, the $p = q$ versus $p \neq q$ split, and deterministic replacement of the two empirical covariances, the complete structural expansion is

$$\begin{aligned}
\mathbb{E}[\Pi_{n,ab}^1] &= D_1 D_2 D_3 \\
&\quad + \frac{D_3}{n} \sum_{u,w} C_2(u, w) \mathbb{E}_1[\phi'(Z_a) \phi'(Z_b) \phi(Z_u) \phi(Z_w)] \\
&\quad + \frac{D_1}{n} \sum_{c,d} C_3(c, d) \mathbb{E}_2[P_{cd}(Z)] \\
&\quad + \frac{1}{n^2} \sum_{c,d,u,w} C_3(c, d) \mathbb{E}_2[\partial_u \partial_w P_{cd}(Z)] \\
&\quad \quad \times \mathbb{E}_1[\phi'(Z_a) \phi'(Z_b) \phi(Z_u) \phi(Z_w)] \\
&\quad + \frac{n-1}{n^2} \sum_{c,d,u,w} C_3(c, d) \mathbb{E}_2[\partial_u L_c(Z)] \mathbb{E}_2[\partial_w R_d(Z)] \\
&\quad \quad \times \mathbb{E}_1[\phi'(Z_a) \phi'(Z_b) \phi(Z_u) \phi(Z_w)] + o(1).
\end{aligned} \quad (\text{B.6})$$

The five displayed branches have orders

$$1, \quad n^{-1}, \quad n^{-1}, \quad n^{-2}, \quad n^{-1}, \quad (\text{B.7})$$

respectively. Thus the leading expectation is

$$\lim_{n \rightarrow \infty} \mathbb{E}[\Pi_{n,ab}^1] = D_1 D_2 D_3. \quad (\text{B.8})$$

The $o(1)$ in (B.6) records deterministic replacement of the empirical group-2 and group-3 covariances. Consequently (B.6) is a complete structural branch expansion after the leading mean-field reduction, not an exact finite-width $1/n$ asymptotic expansion. Fluctuations of those empirical covariances can alter the displayed $1/n$ coefficient even though they cannot alter (B.8).

The final group contains no explicit W^1 factor because the observable is a kernel of derivatives with respect to z^1 . It is peeled by the row law of large numbers

$$\frac{1}{n} \sum_i F(z_i^1(1), \dots, z_i^1(B)) \longrightarrow \mathbb{E}_{Z \sim N(0, G^0)}[F(Z)]. \quad (\text{B.9})$$

Upgrading (B.8) from an expectation limit to convergence of the random kernel requires the separate four-path calculation stated as an obligation in Section 4.4.

Part III. Historical Source Record

Original Mean-Field Peeling Notes and NTK Sample

Status: Frozen historical source record

Conceptual source: Amir Joudaki's original notes and supplied NTK sample

Maintained interpretation: See `.. /CURRENT_RESEARCH_STATE.md` **Checked worked calculations:** See `.. /MUP_TRAINING_CASE_STUDY.md`

This file preserves the original proposal separately from the maintained theory. Spelling and display formatting have been normalized, but the conceptual claims have not been silently corrected. Statements here may be superseded by the maintained documents.

Part I. Original source notes

1. Baseline setup

Use a canonical multilayer perceptron with preactivation

$$z^\ell = W^\ell h^{\ell-1}, \quad h^\ell = \phi(z^\ell), \quad h^0 = x, \quad f(x) = z^L.$$

For a single sample (x, y) , use the loss

$$\mathcal{L}(x, y) = (f(x) - y)^2.$$

Initialization and scaling are intended to follow the μP mean-field feature-learning setup. For convenience, all hidden layers may have width n , and the mean-field limit is $n \rightarrow \infty$.

2. Group ℓ

Suppose z^ℓ is the preactivation generated by the linear weights W^ℓ , with activation ϕ applied to it. Any object that is a W^ℓ term, a z^ℓ term, a $\phi(z^\ell)$ term, a higher activation derivative, or another scalar function of z^ℓ , belongs to group ℓ .

3. Explicit terms

The terms above involve direct weights, preactivations, and postactivations with scalar activation functions such as ϕ , ϕ' , and ϕ'' . These are called explicit group- ℓ terms.

An object such as the backward error

$$\delta^\ell = \frac{\partial \mathcal{L}}{\partial z^\ell}$$

is implicit rather than explicit. Chain rules, product rules, and higher-order derivative rules can expand any finite-order derivative object into a sum of products of explicit terms. The individual products are called product terms.

4. Scalarization

Product terms are normally matrix or higher-order tensor contractions and are therefore noncommutative before their indices are exposed. Any tensor contraction can be written in scalar form by introducing enough indices. For example,

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}.$$

After doing this across a product term, every matrix or tensor product becomes an ordinary scalar product together with explicit index sums. The resulting expressions are scalarized product terms. They involve only scalar weight entries, scalar preactivation entries, and scalar functions of preactivations.

Because the goal is an initialization expectation, linearity allows the expectation to be applied term by term.

5. Grouping and reordering from bottom to top

Within a scalarized product term, scalar multiplication is commutative. Terms can therefore be grouped by layer and ordered from the lowest group index to the highest, with the highest group placed at the far right.

6. Pushing the top-group expectation forward

Suppose the highest active group is m . The intended step is to integrate the fresh randomness of W^m only after all lower-group factors have been placed outside that expectation. Informally, lower-layer randomness is separate from the fresh randomness in W^m .

7. Mean-field multivariate Gaussian limit

The forward pass produces layerwise Gram matrices transported by Gaussian expectations. Schematically,

$$G^{\ell+1} = \mathbb{E}[\phi(X)\phi(Y)^\top], \quad (X, Y) \sim N(0, G^\ell).$$

In the mean-field limit, each neuron row of preactivations is described by a deterministic covariance matrix. Distinct rows are independent. Weight elements are independent unless their indices agree. Weights and same-row preactivations possess a small covariance controlled by the width scaling.

The intended result is that the relevant top-layer weight and preactivation coordinates can be treated through a prescribed joint Gaussian structure.

8. Wick/Isserlis contractions

If a product contains only linear Gaussian terms, Wick's or Isserlis' theorem converts the expectation of the product into a sum over pairings.

9. Stein/Onsager corrections

If nonlinear functions of preactivations occur, Stein's lemma is used to separate the weight-preactivation dependence. Informally, this produces a pseudo-independent contribution plus correction terms involving derivatives of the nonlinear functions and fewer explicit weight factors.

One product term may therefore become a sum of simpler product terms.

10. Weight-activation pseudo-independence terms

Applying the Stein/Onsager step inductively to the highest group is intended to produce terms in which top-layer weights can be integrated separately from top-layer activation-related factors, together with explicit correction branches.

11. Wick contractions on the remaining weight elements

Once the weight dependence has been isolated, Wick/Isserlis contractions remove the remaining weights. Some contraction patterns vanish and others survive, producing Kronecker or Dirac delta constraints among neuron indices.

12. Mean-field peeling

After the top-layer weights have been removed, the remaining nonconstant top-group objects are functions of top-layer preactivations. The original scalarization contains large index sums. Wick pairings break these sums into a small number of equality patterns.

For example,

$$\frac{1}{n^2} \sum_{i,j} \phi(z_i) \phi(z_j) = \left(\frac{1}{n} \sum_i \phi(z_i) \right)^2 \rightarrow (\mathbb{E} \phi(Z))^2,$$

whereas an enforced equality pattern gives

$$\frac{1}{n} \sum_i \phi(z_i)^2 \rightarrow \mathbb{E}[\phi(Z)^2].$$

Each equality pattern defines a family of sums with its own mean-field limit and width scaling. After evaluating these sums, the entire effect of the top group is replaced by scalar Gaussian expectations. All remaining stochastic terms involve lower groups only.

Thus an expectation of scalarized products with highest group m is reduced to a weighted sum of expectations of scalarized products with highest group at most $m - 1$. This is the mean-field peeling step.

13. Repeated peeling

Repeat the same operation one group at a time. Eventually every term is reduced to products of explicit Gaussian expectations of the form

$$\mathbb{E} [f_1(Z_1) f_2(Z_2) \dots f_r(Z_r)],$$

where the covariance of the preactivations is determined by the layerwise forward Gram matrix and the functions are identity, ϕ , ϕ' , ϕ'' , or higher derivatives.

For a fixed batch of size B , the primitive Gaussian integration is intended to be finite-dimensional and controlled by the fixed batch covariance matrices.

14. Practical tips from the original proposal

14.1 Retain only leading width patterns

Only leading surviving width patterns matter for a leading-order mean-field result. A pattern with fewer free assignments may be neglected if it cannot recover the lost width order in later lower-layer summations.

The phrase “contribute similarly downstream” is important: local top-layer counts alone are insufficient if two patterns create different lower-layer sums.

14.2 Recursive scalarization

Full scalarization of every layer at once may be prohibitively complex. Instead, expose and scalarize only the current top group. After it is peeled, repeat on the next group.

14.3 Recursive states and dynamic programming

The equality patterns and lower-layer boundary signatures created by one peel may close on a finite family of states. If so, the computation can be written as a recursive or dynamic-programming system.

15. Original headline claim

The original intended headline was:

Any object involving derivatives or higher-order derivatives at initialization of a depth- L MLP in the mean-field regime, with arbitrary but constant $O(1)$ maximum derivative order and a fixed batch of B samples, can be computed through a set of $O(L)$ recursive equations involving only explicit Gaussian expectations of layerwise preactivations passed through identity, ϕ , ϕ' , ϕ'' , and higher derivatives.

16. Original feature-learning motivation

Under a μP mean-field feature-learning parameterization, the change in a hidden-layer Gram matrix after one step can be expanded to fixed order in the learning rate using higher derivatives of the network at initialization. At later steps, gradient corrections can similarly be expanded using higher derivatives.

In the gradient-flow scaling

$$t \rightarrow \infty, \quad \eta \rightarrow 0, \quad t\eta = \tau,$$

the coefficients become time derivatives of the training trajectory at initialization. Each such coefficient is intended to be an object handled by the peeling program.

17. Original conjecture

For a neural network evaluated on fixed inputs and possessing a well-defined mean-field learning limit, any scalar observable such as loss or a hidden-layer Gram matrix, provided its training curve is locally smooth, has initialization derivatives expressible as sums of products of explicit Gaussian integrals. Their covariance matrices are finite and determined by the input, output, architecture, initialization, and loss configuration.

18. Role of the original NTK sample

The supplied LaTeX note attempted a first-principles NTK derivation using forward covariance recursions, backward Jacobian moments, Wick contractions, and Stein corrections. Its intended lessons were:

- forward preactivation covariances obey a deterministic Gaussian recursion;
- backward influence can be propagated recursively;
- the classical NTK is built from forward activation kernels and derivative-transport factors;
- diagonal and off-diagonal backward patterns must be distinguished;
- off-diagonal corrections should be checked rather than discarded by an independence assumption.

The sample is a source of intuition, not an adopted proof. Its notation changes weight-index conventions, switches from square loss to a linear-output surrogate, and leaves several second-moment and off-diagonal states incompletely defined.

Part II. Original NTK sample execution

A.1 Title and stated purpose

Neural Tangent Kernel: A First Principles Derivation

Amir Joudaki, ETH Zurich

The sample described itself as a first-principles derivation of the Neural Tangent Kernel using the law of large numbers, central limit theorem, and Gaussian integration identities. It aimed to establish mean-field variance recursions, analyze kernel shifts through parameter updates, derive the classical NTK recursion, analyze possible off-diagonal backward interactions, and argue that these corrections are negligible at infinite width.

A.2 Network architecture in the sample

The sample used an input layer, $L - 1$ hidden layers, and a scalar output:

$$h^0(x) = x \in \mathbb{R}^{d_0},$$

$$z^\ell(x) = W^{\ell-1} h^{\ell-1}(x) + b^{\ell-1},$$

$$h^\ell(x) = \phi(z^\ell(x)), \quad 1 \leq \ell < L,$$

and

$$z^L(x) = W^{L-1} h^{L-1}(x) + b^{L-1}.$$

Weights and biases were initialized as

$$W_{ij}^\ell \sim N\left(0, \frac{\sigma_w^2}{d_\ell}\right), \quad b_i^\ell \sim N(0, \sigma_b^2).$$

For the sample derivation, the “loss” was changed from the squared loss in the baseline notes to the scalar output

$$\mathcal{L} = z^L(x').$$

The stated motivation was to isolate the gradient structure and obtain a simple first-moment boundary.

A.3 Forward kernel recursion in the sample

The preactivation kernel was defined by

$$\Sigma^\ell(x, x') := \frac{1}{d_\ell} \langle z^\ell(x), z^\ell(x') \rangle.$$

The proposed infinite-width recursion was

$$\Sigma^0(x, x') := \sigma_w^2 \frac{x^\top x'}{d_0} + \sigma_b^2,$$

$$\Sigma^{\ell+1}(x, x') := \sigma_w^2 \mathbb{E}[\phi(z)\phi(z')] + \sigma_b^2,$$

where (z, z') is centered Gaussian with covariance

$$\begin{pmatrix} \Sigma^\ell(x, x) & \Sigma^\ell(x, x') \\ \Sigma^\ell(x, x') & \Sigma^\ell(x', x') \end{pmatrix}.$$

The derivative kernel was defined as

$$\dot{\Sigma}^\ell(x, x') := \sigma_w^2 \mathbb{E}[\phi'(z)\phi'(z')],$$

with the Gaussian covariance inherited from the previous forward layer.

A.4 Influence function in the sample

For a parameter block W^{k-1} , the sample defined the influence on layer ℓ by

$$\dot{\Theta}_{k,\ell} := \frac{1}{d_\ell} \sum_{i=1}^{d_\ell} \left\langle \frac{\partial z_i^\ell(x)}{\partial W^{k-1}}, \frac{\partial z^L(x')}{\partial W^{k-1}} \right\rangle_{W^{k-1}}.$$

Using the derivative of Wa , it rewrote this schematically as

$$\dot{\Theta}_{k,\ell} = \langle h^{k-1}(x), h^{k-1}(x') \rangle \frac{1}{d_\ell} \sum_i \left\langle \frac{\partial z_i^\ell(x)}{\partial z^k}, \frac{\partial z^L(x')}{\partial z^k} \right\rangle_{z^k}.$$

A.5 Backward Jacobian moments in the sample

With

$$J_j^{(\ell \leftarrow k)}(x) := \frac{\partial z^\ell(x)}{\partial z_j^k},$$

the sample introduced

$$M_{k,\ell} := \frac{1}{d_\ell} \sum_{i=1}^{d_\ell} \mathbb{E} \left[\sum_{j=1}^{d_k} J_{j,i}^{(\ell \leftarrow k)}(x) J_j^{(L \leftarrow k)}(x') \right],$$

$$Q_{k,\ell} := \frac{1}{d_\ell} \sum_{i=1}^{d_\ell} \mathbb{E} \left[\left(\sum_{j=1}^{d_k} J_{j,i}^{(\ell \leftarrow k)}(x) J_j^{(L \leftarrow k)}(x') \right)^2 \right],$$

and

$$V_{k,\ell} = Q_{k,\ell} - M_{k,\ell}^2.$$

A.6 First-moment recursion proposed in the sample

The sample stated

$$M_{k,\ell} = \dot{\Sigma}^k(x, x') M_{k+1,\ell},$$

and therefore

$$M_{k,\ell} = M_{\ell,\ell} \prod_{i=k}^{\ell-1} \dot{\Sigma}^i(x, x').$$

Its proof expanded the Jacobian recursion

$$\frac{\partial z_i^{\ell+1}}{\partial z_j^\ell} = W_{ij}^\ell \phi'(z_j^\ell)$$

inside $M_{k,\ell}$, producing a term of the form

$$M_{k,\ell} = \frac{1}{d_\ell} \sum_{i,j,s,r} \mathbb{E} \left[\frac{\partial z_i^\ell(x)}{\partial z_s^{k+1}} \frac{\partial z^L(x')}{\partial z_r^{k+1}} \right. \\ \left. \times \phi'(z_j^k(x)) \phi'(z_j^k(x')) W_{sj}^k W_{rj}^k \right].$$

The source proof then invoked independence of the backward Jacobians from W^k and used

$$\mathbb{E}[W_{sj}^k W_{rj}^k] = \delta_{sr} \frac{\sigma_w^2}{d_k}.$$

It concluded with a derivative-kernel factor times the next backward moment. The maintained theory does not adopt this independence step without a Wick–Stein analysis, because upper activation derivatives can contain preactivations depending on W^k .

A.7 Boundary conditions proposed in the sample

The sample stated

$$M_{\ell,\ell} = \begin{cases} 0, & \ell < L, \\ 1, & \ell = L. \end{cases}$$

The reasoning was that

$$M_{\ell,\ell} = \frac{1}{d_\ell} \sum_i \mathbb{E} \left[\frac{\partial z^L(x')}{\partial z_i^\ell} \right],$$

which equals one at the scalar output and was asserted to vanish at hidden layers by independence and symmetry. The resulting proposed complete first moment was

$$M_{k,\ell} = \begin{cases} 0, & \ell < L, \\ \prod_{i=k}^{L-1} \dot{\Sigma}^i(x, x'), & \ell = L. \end{cases}$$

A.8 Second-moment statement in the sample

The sample proposed

$$Q_{k,\ell} = \left(\dot{\Sigma}^k(x, x') \right)^2 Q_{k+1,\ell} + O(1/d_k),$$

with boundary relation

$$Q_{\ell,\ell} = Q_{\ell+1,\ell+1} \dot{\Sigma}^{\ell+1}(x', x') \frac{d_{\ell+1}}{d_\ell}.$$

The proof was described as following from an analogous Wick expansion, with the leading pairing producing the squared derivative-kernel term. The source did not completely define or prove all normalization and boundary details.

A.9 NTK expression in the sample

For a learning-rate scaling of the form $\eta_k \propto \eta_0/d_{k-1}$, the source proposed the expected layer contribution

$$\mathbb{E}[\dot{\Theta}_{k,L}] = \eta_0 \Sigma^{k-1}(x, x') \prod_{i=k}^{L-1} \dot{\Sigma}^i(x, x').$$

Summing parameter blocks gave

$$\sum_{k=1}^L \mathbb{E}[\dot{\Theta}_{k,L}] = \eta_0 \sum_{k=1}^L \Sigma^{k-1}(x, x') \prod_{i=k}^{L-1} \dot{\Sigma}^i(x, x').$$

The NTK recursion was then stated as

$$\Theta^L(x, x') = \Theta^{L-1}(x, x') \dot{\Sigma}^{L-1}(x, x') + \Sigma^{L-1}(x, x'),$$

with base case

$$\Theta^1(x, x') = \Sigma^0(x, x').$$

A.10 Signal propagation remark

For $x = x'$, the source wrote

$$\dot{\Sigma}^\ell(x, x) = \sigma_w^2 \mathbb{E}[\phi'(Z)^2],$$

and observed that the product of derivative kernels controls backward signal propagation. It identified the edge-of-chaos condition with

$$\chi_1 := \sigma_w^2 \mathbb{E}[\phi'(\sqrt{q^\star} Z)^2] \approx 1.$$

A.11 Extended Stein identities in the sample

For jointly Gaussian (X, Y, Z) , the source recorded identities of the form

$$\mathbb{E}[f(X)Y] = \sigma_{XY} \mathbb{E}[f'(X)],$$

and

$$\mathbb{E}[f(X)f(Y)Z] = \sigma_{XZ} \mathbb{E}[f'(X)f(Y)] + \sigma_{YZ} \mathbb{E}[f(X)f'(Y)].$$

These were intended to control forward-backward dependence missed by naive independence.

A.12 Tentative diagonal/off-diagonal system

The source proposed a coupled system

$$D_{k,\ell} = \dot{\Sigma}^{k+1} D_{k+1,\ell} + \frac{\sigma_w^4}{d_k} (\mu_{\phi''}^{k+1})^2 C_{\phi,\phi}^k S_{k+1,\ell},$$

$$S_{k,\ell} = 2\sigma_w^4 C_{\phi'',\phi''}^{k+1} (\mu_{\phi}^{k+1})^2 D_{k+1,\ell} + (\mu_{\phi}^{k+1})^2 (\mu_{\phi''}^{k+1})^2 S_{k+1,\ell}.$$

Here D was intended to represent diagonal backward contributions and S off-diagonal corrections. The source claimed:

1. off-diagonal corrections are $O(1/d_k)$ or carry activation-mean factors;
2. centered activations make some off-diagonal terms vanish;
3. the classical NTK is recovered at leading order.

These equations remain archival and tentative because their normalization, state definitions, and boundary conditions were not fully specified.

A.13 Conclusions stated in the sample

The sample concluded that:

1. infinite width produces clean law-of-large-numbers recursions;
2. only the last layer contributes to the chosen first moment under the linear-output surrogate loss;
3. variance contributions vanish at infinite width;
4. off-diagonal backward interactions are negligible;
5. the NTK emerges from parameter-gradient inner products.

It also emphasized that the NTK is independent of the learning rate as an architectural kernel, while the learning rate enters the training update

$$\Delta f = -\eta \Theta \partial_f \mathcal{L}.$$

Finally, it suggested extension to residual networks, normalization layers, and other architectures by modifying the forward and backward recursions.

A.14 Current archival status

This source execution remains important because it supplied the motivating example for the peeling program. It should not be cited as a completed proof without repairing:

- the inconsistent layer indexing;
- the change of loss;
- raw versus effective weight normalization;
- the dependence of upper backward Jacobians on the current weight layer;
- the definitions and normalization of the second-moment states;
- the tentative off-diagonal recursion.

The checked backward-kernel execution in the maintained case study is the current replacement for the corresponding independence-based step.

Source-history note

The maintained living document was first consolidated on 2026-08-01. The μP training case study was added on 2026-08-03. During those revisions the global-Gaussian and pseudo-independence language was replaced by exact layerwise conditional Gaussianity and Wick–Stein partial matchings; scalar contraction states, equality partitions, width power counting, deterministic Gram replacement, feature-learning jets, and explicit proof obligations were introduced. Those revisions now live in the two maintained Markdown sources, not in this frozen source record.